

Design, Simulation, Testing, and Hardware Validation of an LTC1871-Based DC-DC Boost Converter for Battery- Powered Subsystems for Robotic & Autonomous Devices.

LTC1871-Based DC-DC Boost Converters

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Abstract

Autonomous ground vehicles require stable, efficiently regulated power rails to ensure reliable operation of sensitive payload electronics throughout a full battery discharge cycle. This report presents the design, simulation, verification, and hardware implementation of a DC-to-DC boost converter for battery-powered robotic and autonomous devices. The converter, based on the Linear Technology LTC1871 current-mode controller, accepts a three-cell lithium polymer battery input ranging from 8–11 V and regulates a stable 12 V output at up to 2 A continuous load current. The project followed a structured methodology beginning with analytical power stage design, during which an initial converter topology was abandoned due to simulation convergence failures and replaced with a boost topology confirmed by sufficiency analysis. A comprehensive simulation campaign validated all performance metrics with substantial margin across the full input range, including an output voltage of 12.20V, output ripple of 171 mV, load transient droop of 123mV under a 1.5A inrush step, and a simulated efficiency of 89.84% at the upper input boundary. A three-layer input protection scheme was developed comprising a series P-channel transistor for reverse polarity blocking, a resistive divider for undervoltage lockout, and a Zener-transistor clamp for overvoltage lockout, together forming a voltage window that restricts active regulation to a validated operating band of 8.0 to 11.1V. Iterative tuning of the divider ratio and clamp network was required to counteract parasitic loading effects that initially compressed the operating window, with the final design achieving precise alignment with the target battery discharge range. The validated schematic was translated into a compact two-layer printed circuit board measuring 60.5 by 42.5 millimetres using an open-source layout tool, with fabrication files generated for external manufacture and final assembly carried out by hand-soldering in the university electronics laboratory. Preliminary bench testing confirmed the UVLO divider operation to within 0.42 % of its designed value; however, the controller IC was permanently damaged during oscilloscope probing of the fine-pitch package, preventing full electrical characterisation.

Declaration

I hereby certify that this report and the project that it describes is my original work and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged in the report. I would like to thank my supervisor, **Dr. Bob Lawlor**, for his guidance and feedback throughout the project. I also wish to thank **Mr. John Maloco**, Electronics Laboratory Technician, for his assistance with component procurement and hand-soldering of the printed circuit board. I confirm that any use of Generative AI tools in the preparation of this report complies with Maynooth University's guidelines. Where GenAI tools have been used to support my work, I have acknowledged this appropriately. A list of prompts is provided in the appendix.

Signed:



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1 Introduction

Mobile robotics platforms depend on compact battery packs whose terminal voltage inevitably sags as the cells discharge, a 3-cell lithium-polymer pack, for instance, falls from 12.6V fully charged to below 9V at end-of-discharge, easily dropping below the voltage required to keep downstream point-of-load regulators in regulation [1,2]. All three payload devices in this project, the Raspberry Pi Compute Module 4 [20], the RPLiDAR A1 laser scanner, and the RFD900x telemetry modem, are fundamentally 5V loads with tight input tolerances. The RPLiDAR's core scanner accepts only 4.9-5.5V and is damaged above 5.5V, while the RFD900x specifies a +5V nominal supply with an absolute maximum of +6V. [3, 4] Neither device can be powered directly from a fluctuating battery rail. Instead, the system employs a cascaded power architecture: a tightly regulated 12V intermediate bus first is generated from the battery, and each subsystem is then supplied by a dedicated point-of-load step-down converter that delivers a precise 5V. If the 12V bus sags below the dropout threshold of these downstream converters, their output collapses and the loads brown out, regardless of remaining battery capacity. A stable 12V backbone, maintained across the entire discharge curve, therefore guarantees the headroom needed for the 5V conversion stages and ensures continued operation of the LiDAR, telemetry link, and processor throughout an autonomous mission [5, 6].

1.1 Problem Statement and Scope

The specific engineering problem addressed in this thesis is the design and implementation of a regulated intermediate power bus for the payload of a proof-of-concept autonomous ground vehicle. The bus is generated from a 3-cell lithium-polymer battery whose terminal voltage during operation is intentionally restricted to an 8–11V window, below the 12V target, so that a boost-only topology can be employed. The converter is built around the LTC1871 current-mode boost controller, which steps up the battery voltage to a tightly regulated 12V rail while supplying the three 5V payload devices (Raspberry Pi Compute Module 4, RPLiDAR A1, RFD900x modem) through their downstream point-of-load regulators [7, 8]. The converter is rated to deliver up to 24W at the full 2A continuous load, providing sufficient margin for the combined power draw of the three payload subsystems and their associated point-of-load conversion losses. The scope of this work is confined to the boost converter itself: the design, simulation, and hardware prototyping of the power stage, compensation network, and feedback loop. The downstream 5V regulators are treated as fixed resistive loads, and all navigation, sensing, and communication algorithms lie outside the project boundary. The converter's performance is evaluated against output voltage regulation, efficiency, load transient response,

and the ability to maintain a stable 12V rail across the full 8–11V input range, evaluated primarily through LTspice simulation, with hardware prototyping conducted to validate the physical design. By constraining the problem to this single boost stage, the thesis isolates the power-conditioning function that enables the robot's sensitive electronics to survive a complete battery discharge cycle without supply interruption.

1.2 Approach and Methodology

An initial SEPIC design using the TPS61088 controller was abandoned due to persistent LTspice convergence failures [9, 10]. A topology sufficiency analysis confirmed that a simple boost converter could meet the specification because the battery's 8–11 V operating window lies strictly below the 12 V output. The design was migrated to the LTC1871 current-mode boost controller, and the power stage was sized through first-principles calculations: switching frequency, duty-cycle range, inductor and output capacitor selection (including DC-bias derating), MOSFET and Schottky diode choices, feedback divider derivation, and Type II compensation network design for loop stability [11].

The analytical design was subjected to a structured LTspice transient simulation campaign that validated eight performance metrics across the full input range, supplemented by analytical loop stability verification in MATLAB [23]. The validated design was translated into hardware with a three-layer input protection scheme (reverse polarity blocking via a DMP3099L PMOS, undervoltage lockout, and overvoltage lockout). The protection thresholds were iteratively tuned through simulation to yield an 8.0–11.1 V operating window matching the battery specification [12]. The final schematic was captured in KiCad, laid out as a two-layer PCB, fabricated by JLCPCB, and hand-assembled in the university laboratory [13, 14].

1.3 Evaluation Strategy

The design was validated primarily through LTspice transient simulations and analytical loop stability verification in MATLAB, with additional verification provided by a fabricated hardware prototype. Transient simulations were used to evaluate output voltage regulation, ripple, line and load regulation, transient response, peak inductor current, and efficiency across the full input voltage range. AC analysis was performed to confirm loop stability through extraction of phase and gain margins. Input protection functionality was verified by sweeping the input voltage across the expected operating regions. A fabricated PCB confirmed schematic-to-layout translation and manufacturing feasibility. Detailed simulation test conditions, measurement procedures, and results are presented in Chapters 4 and 5. Electrical bench testing, thermal imaging, EMI measurements, and long-term reliability testing were not performed; these are

noted as future work. Pre-hardware simulation validation is standard practice in power converter development, as LTspice's device-level models – including the manufacturer-supplied LTC1871 macro model – capture the dominant electrical behaviour with sufficient fidelity to confirm design feasibility before committing to fabrication.

1.4 Broader Impact

This project provides a regulated 12 V rail from a 3-cell Li-Po battery pack (8.0 – 11.1V) for autonomous ground vehicle applications such as environmental monitoring, agricultural inspection, or search and rescue. The converter achieves 89 % efficiency at full load, dissipating approximately 2.7 W of heat; an equivalent design operating at only 80 % efficiency would waste 6 W, nearly doubling thermal losses. This improvement directly supports SDG 7 (Affordable and Clean Energy) by extending battery life and reducing charging cycles, which lowers net energy consumption over the vehicle's operational lifetime. The higher efficiency translates into a measurable increase in mission duration compared to a conventional 80 %-efficient converter, particularly for subsystems that draw heavily from the 12 V rail. This enables longer-lasting field deployments.

The design also contributes to SDG 9 (Industry, Innovation and Infrastructure) by demonstrating a robust, reproducible DC-DC converter architecture suitable for battery-powered robotic systems, and to SDG 12 (Responsible Consumption and Production) through the use of lead-free, standard discrete components that facilitate recycling and reduce hazardous waste [15–17]. No inherent ethical conflicts are associated with this design, as it is intended for general-purpose power conversion on autonomous platforms.

1.5 Original Contributions and Externally Provided Resources

The following original contributions were made by the author:

- Design and sizing of the power stage (inductor, MOSFET, diode, output capacitors) for a 0.5A to 2A load range.
- Selection and configuration of the compensation network to achieve stable loop response.
- Implementation of input window protection (UVLO/OVLO) using a resistive divider and Zener-NPN clamp circuit.
- Full LTspice simulation framework, including parametric sweeps, transient measurement windows, and Bode analysis.

- PCB layout, component sourcing, and assembly oversight.

Externally provided resources are acknowledged as follows:

- Dr. Lawlor provided supervisory guidance on converter topology selection and review of the feedback loop design.
- John Maloco performed the soldering of surface-mount components on the fabricated PCB.
- Third-party SPICE models for the LTC1871 controller, MOSFETs, and diode were sourced from the manufacturer (Analog Devices) and third-party libraries; their limitations are noted in the simulation methodology.
- JLCPCB fabricated the bare PCB according to the submitted Gerber files.

1.6 Structure of the Report

Chapter 2 reviews the relevant background theory of boost converters, including continuous-conduction mode, feedback compensation, and input protection principles. Chapter 3 presents the design methodology, covering component selection, loop compensation calculations, and protection threshold setting. Chapter 4 describes the simulation setup, parametric test conditions, and the metrics used for validation. Chapter 5 reports the results from transient simulations and analytical loop stability verification, including efficiency, ripple, load transient response, and protection verification. Chapter 6 discusses the limitations of the simulation-only validation and the status of the hardware prototype, followed by conclusions and recommendations for future work.

2 Technical Background

This chapter establishes the theoretical foundation for the converter design. The key specifications driving the analysis are: an 8–11 V input from a 3S Li-Po battery, a regulated 12 V output at 2 A continuous load, 350 kHz switching frequency, and current-mode control via the LTC1871 controller. The chapter covers DC–DC converter fundamentals, boost converter operating principles, current-mode control theory, feedback loop compensation, input protection topologies, and PCB layout considerations.

2.1 DC-DC Converter Fundamentals

A battery delivers a variable, unregulated voltage, yet the load requires a fixed, stable potential. In robotic systems, loads are time-varying: motors draw inrush currents during setup, LiDAR's

and communication radios exhibit burst-mode power demands, and compute modules require clean rails to avoid erratic behaviour. Poor regulation under these transients can cause brownouts, sensor data corruption, or radio dropouts. Consequently, a regulator must be placed between the source and the load. A linear regulator (e.g., LM7812) can only step down, dissipating the voltage difference as heat; with an 8-11V input and a 12V output, a conventional linear regulator cannot step up, and even in step-down applications its efficiency is limited to $\frac{V_{out}}{V_{in}}$. In contrast, a switch-mode converter stores energy in an inductor during one part of a switching cycle and releases it to the load during another, achieving typical efficiencies of 85-95%.

Three fundamental topologies exist in continuous conduction mode (CCM). The buck converter produces an output voltage lower than the input, with the transfer function:

$$\frac{V_{out}}{V_{in}} = D \quad (1)$$

Where D is the duty cycle. The boost converter steps up the voltage, following:

$$\frac{V_{out}}{V_{in}} = \frac{1}{(1 - D)} \quad (2)$$

The basic buck-boost converter can either step up or step down, but it inverts the output polarity:

$$\frac{V_{out}}{V_{in}} = -\frac{D}{(1 - D)} \quad (3)$$

For applications requiring a non-inverting buck-boost capability, the SEPIC (single-ended primary inductance converter) is a practical alternative, while preserving polarity.

Given the specification of an 8-11V input (from a 3S Li-Po battery) and a regulated 12V output, the input voltage always lies below the output voltage. Thus, a simple boost topology is sufficient; the additional flexibility of a buck-boost or SEPIC converter would introduce unnecessary component and control complexity. This choice motivated the detailed analysis of boost converter operation presented in Section 2.2.

2.2 Boost Converter Operating Principle

Figure 2.1 shows the canonical boost converter circuit, consisting of four principal elements: an inductor L, a power MOSFET switch Q_1 , a diode D_1 , and an output capacitor C_{out} . The input voltage source V_{in} (here, a 3S Li-Po battery) supplies energy to the inductor, while the output

delivers a regulated higher voltage to the load R_{load} . The LTC1871 controller drives the MOSFET gate; a current-sense signal is derived from the MOSFET drain-source voltage, eliminating the need for a discrete sense resistor.

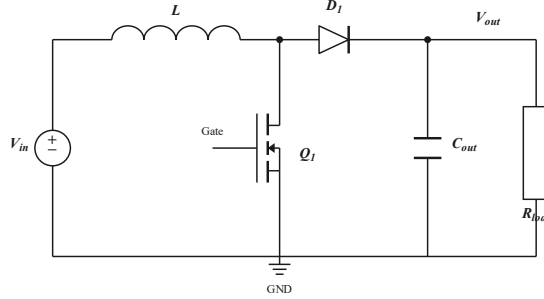


Figure 1: Boost converter topology - inductor L , MOSFET $Q1$, diode $D1$, output capacitor C_{OUT} , input voltage V_{IN} , and load

The converter operates in two distinct intervals per switching cycle. During the on interval ($0 < t < T_s$), the MOSFET conducts, connecting the inductor-diode junction to ground. The diode becomes reverse-biased, disconnecting the input from the output. The inductor voltage equals V_{in} , causing it's current to ramp linearly upward, storing energy. During this interval, the output capacitor solely supplies the load current, and the capacitor voltage decreases slightly.

During the off interval ($DT_s < t < T_s$), the MOSFET turns off. The inductor's magnetic field collapses. Reversing the polarity across the inductor such that its voltage adds to V_{in} . The diode becomes forward biased, and the inductor current flows through the diode into the output capacitor and load. The inductor voltage now equals $V_{in} - V_{out}$ (negative, since $V_{out} > V_{in}$), so the inductor current ramps downward, transferring the stored energy to the output.

In steady state, the net change in inductor current over one switching cycle must be zero – the volt-second balance principle. Applying this to the boost converter:

- On interval ($0 < t < DT$): $v_L = V_{IN}$
- Off interval ($DT < t < T$): $v_L = V_{IN} - V_{OUT}$

Expanding and simplifying:

$$V_{in}D + V_{in}(1 - D) - V_{out}(1 - D) = V_{in} - V_{out}(1 - D) = 0 \quad (4)$$

Solving for the voltage transfer ratio:

$$\frac{V_{out}}{V_{in}} = \frac{1}{1 - D} \text{ and } D = 1 - \frac{V_{in}}{V_{out}} \quad (5)$$

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For the design, $V_{out} = 12V$. With $V_{in} = 11V$ (battery near full), the required duty cycle is $D = 1 - \frac{11}{12} = 0.083$ (8.3%).

At the minimum input $V_{in} = 8V$, $D = 1 - \frac{8}{12} = 0.333$ (33.3%). All duty cycles remain below 50%, so the LTC1871's internal slope compensation circuit remains inactive, simplifying loop dynamics.

During the on interval, the inductor voltage is V_{in} . From the inductor voltage-current relationship:

$$V = L \frac{di}{dt} \quad (6)$$

Shifting the terms around and make di the subject of the formula:

$$\frac{V}{L} = \frac{di}{dt} \quad (7)$$

During the on interval the inductor voltage is V_{IN} :

$$\Delta I_L = \frac{V_{in} D}{f_{sw} L} \quad (8)$$

The average inductor current is derived from power balance, assuming ideal efficiency ($P_{in} = P_{out}$): $V_{in} I_{L,AVG} = V_{OUT} I_{OUT}$. Substituting $\frac{V_{OUT}}{V_{IN}} = \frac{1}{(1-D)}$ gives:

$$I_{L,AVG} = \frac{I_{OUT}}{1 - D} \quad (9)$$

This analytical peak is lower than the LTC1871's internal current-limit threshold (nominally 3.59A for this design); the inductor's saturation rating must exceed the higher of the two. The selected inductor (XAL6060-103MEC) has a saturation current rating of 11.2A, providing ample margin.

The inductor current waveform in CCM consists of a DC average $I_{L,avg}$ with a superimposed triangular ripple ΔI_L (peak-to-peak). The RMS (root-mean-square) current is needed to calculate the copper loss in the inductor winding and to verify that the selected inductor does not exceed its thermal current rating. For a triangular ripple on a DC offset, the RMS current is given by:

$$I_{L,RMS} = \sqrt{I_{L,avg}^2 + \frac{\Delta I_L^2}{12}} \quad (10)$$

The RMS current is the quantity that determines the inductor's copper loss: $P_{Cu} = I_{L,RMS}^2 \cdot DCR$, where DCR is the winding DC resistance. The three current metrics therefore serve distinct purposes:

- **Average current** $I_{L,avg}$ — used in power balance and output current delivery.
- **Peak current** $I_{L,pk}$ — used to ensure the inductor does not saturate (must be below I_{sat}).
- **RMS current** $I_{L,RMS}$ — used to ensure the inductor does not overheat (must be below the manufacturer's RMS current rating).

The selected inductor (XAL6060-103MEC) has an RMS current rating of 8.3 A, providing a large margin over the 3.01 A calculated above.

During the MOSFET on interval, the output capacitor alone supplies the load current. The charge removed is:

$$\Delta Q = \frac{I_{OUT}D}{f_{sw}} \quad (11)$$

Giving a charge-based ripple of:

$$\Delta V_{OUT, ESR} = \Delta I_L R_{ESR} \quad (12)$$

The total output voltage is the sum of the charge-based and ESR contributions. Furthermore, the output capacitor must be rated to carry the AC ripple without excessive heating. For a boost converter in CCM, the RMS ripple current through the output capacitor is:

$$I_{RMS(C_{out})} = I_{OUT} \sqrt{\frac{V_{OUT} - V_{IN}}{V_{IN}}} \quad (13)$$

Finally, for ceramic (MLCC) capacitors, the effective capacitance decreases under DC bias. Manufacturer derating curves must be consulted; the design should oversize the nominal capacitance accordingly. The detailed selection and numerical verification are presented in Chapter 3.

The input capacitor handles the AC component of the inductor current. In a boost converter, the input current is continuous, and only the ripple portion flows through C_{in} . The RMS ripple current is:

$$I_{RMS(C_{in})} = \frac{\Delta I_L}{\sqrt{12}} \quad (14)$$

The input voltage ripple has a capacitive component and an ESR component. Excessive input ripple can degrade the accuracy of the LTC1871's $R_{DS(on)}$ -based current sensing and may perturb the UVLO divider threshold. Therefore, adequate input capacitance (typically a low-ESR ceramic placed directly at the IC's VIN pin) is required. The specific capacitance value, derating considerations, and verification of negligible ripple are provided in Chapter 3.

2.3 Current-Mode Control and the LTC1871 Architecture

Two approaches exist for controlling a switch-mode converter. In voltage-mode control, the error amplifier directly sets the duty cycle; the power stage exhibits a second-order LC response requiring a Type III compensator. In current-mode control, an inner current loop terminates each switching cycle when the inductor current reaches a threshold set by the outer voltage loop.

Current-mode control, by contrast, introduces an inner current loop. The outer voltage error amplifier sets a peak-current threshold, and the inner loop terminates each switching cycle when the inductor current reaches that threshold. This converts the inductor from a voltage-driven element (which contributes a second-order response) into a controlled current source (first-order, single-pole response). The resulting plant behaves approximately as a single pole at the output capacitor's load resistance, dramatically simplifying compensation. The LTC1871 controller implements current-mode control using the MOSFET's on-resistance for sensing, as described in its datasheet. [19]

The architecture consists of two nested feedback loops. The inner current loop operates cycle-by-cycle at the switching frequency (350kHz). The outer voltage loop operates through the compensation network connected to the ITH pin, with a crossover frequency of 4.91kHz (as designed in Section 4.6). The ITH pin serves as the interface between the two loops: the outer loop sets the ITH voltage, which the inner loop uses as a peak-current reference. The inner loop responds to input voltage perturbations within one switching cycle ($2.86\mu s$), before the output capacitors can charge or discharge meaningfully, a mechanism that produces the line regulation of $2.9\mu V/V$ measured in simulation (Chapter 5)

The LTC1871 eliminates the need for a discrete current-sense resistor by using the MOSFET's drain, and the internal current-sense comparator measures the voltage drop across $R_{DS(on)}$ during the on-interval. This lossless sensing maximises efficiency but introduces a temperature dependency because $R_{DS(on)}$ increases with junction temperature. For this design, the duty cycle

range is 8.3% to 33.3%, and the selected MOSFET (IRF7811) has a typical $R_{DS(on)}$ of 12m Ω . The resulting sense voltage remains well within the LTC1871's current-limit threshold across the expected temperature range, so no external compensation for temperature is required.

Slope compensation is a standard addition to current-mode controllers to prevent subharmonic oscillation at duty cycles above 50%. The LTC1871 includes internal slope compensation that activates automatically when the duty cycle exceeds 50%. Because the present design operates entirely below 50% duty cycle ($D_{MAX} = 33.3\%$), the internal slope compensation remains inactive. This simplifies the loop dynamics and further reduces the component count; it is a direct benefit of the boost-only topology section (Chapter 5).

For this battery-powered robotic application, current-mode control offers several advantages over voltage-mode control: the power stage becomes first-order (requiring only a Type II compensator), peak-current limiting provides inherent cycle-by-cycle overcurrent protection, input voltage variations are rejected within one switching cycle, and the LC double pole is eliminated from the control-to-output transfer function. These characteristics make the LTC1871 well-suited to the variable input voltage and transient load demands of mobile robotic systems [18], [19].

2.4 Feedback and Loop Compensation

The feedback loop maintains tight output voltage regulation despite input voltage and load current variations. The loop gain determines both steady-state accuracy and dynamic performance. The crossover frequency must remain below the right-half-plane zero (RHPZ), a boost-specific constraint where increasing duty cycle momentarily reduces output current, causing a phase inversion that cannot be compensated.

The output voltage is sensed by a resistive divider connected to the FB pin of the LTC1871. The internal reference voltage is $V_{REF} = 1.230\text{ V}$. In steady state, the error amplifier forces the FB pin to equal V_{REF} . The relationship between the output voltage and the divider resistors is given by:

$$V_{OUT} = 1 + \frac{R_{TOP}}{R_{BOT}} \quad (15)$$

Where R_{TOP} is the resistor from V_{OUT} to FB and R_{BOT} is the resistor from FB to ground. The absolute values of these resistors are selected to balance noise immunity (favoured by lower impedance) against quiescent current consumption (favoured by higher impedance). This divider constitutes the sensing element of the outer voltage loop; its accuracy directly influences output voltage regulation.

The loop gain $T(s) = G_{plant}(s) \cdot H(s) \cdot G_c(s)$ describes the propagation of a perturbation around the feedback path and is analysed using Bode plots of magnitude and phase. Three key metrics determine stability and performance:

- **Crossover frequency f_c** : the frequency at which $|T(f_c)| = 1$ (0dB). This provides an indication of closed loop bandwidth.
- **Phase Margin**: the difference between the phase of $T(s)$ at f_c and -180° . Values below 45° risk ringing or instability.
- **Gain Margin**: the amount by which $|T(s)|$ is below 0 dB at the frequency where phase equals -180° . A margin of at least 10dB is generally desired.

A fundamental limitation in boost converters is the Right-Half-Plane-Zero (RHPZ), which arises because an increase in duty cycle initially reduces the energy delivered to the output before the inductor stores additional energy. Its location is approximately:

$$f_{RHPZ} = \frac{(1-D)^2 R_{OUT}}{2\pi L} \quad (16)$$

Because the RHPZ contributes gain but negative phase shift, the crossover frequency must be placed significantly below f_{RHPZ} (commonly $f_c \leq \frac{f_{RHPZ}}{5}$) to preserve stability. This constraint is evaluated at the worst-case operating point.

Owing to the single-pole behaviour of the current-mode controlled power stage, a Type II compensator suffices. The network comprises a series resistor R_z and capacitor C_z from the ITH pin to ground, together with a smaller capacitor C_p . The compensator introduces:

- A zero at $f_z = \frac{1}{2\pi R_z C_z}$, which provides phase boost near the crossover frequency.
- A high-frequency pole at $f_p = \frac{1}{2\pi R_z C_p}$, which attenuates switching noise.
- An integrator pole at the origin arising from the error amplifier, ensuring zero steady-state error.

Collectively, these elements shape the loop response to achieve the desired crossover frequency with adequate phase margin while maintaining low output impedance and fast transient response.

2.5 Input Protection Topologies

Battery-powered robotic platforms are exposed to reverse battery connection, undervoltage during deep discharge, and overvoltage transients. Three protection mechanisms are implemented: a series P-channel MOSFET for reverse polarity blocking, a resistive divider for undervoltage lockout (UVLO), and a Zener-NPN clamp for overvoltage lockout (OVLO). Together these form a voltage window that enables the converter only within the safe 8.0–11.1 V operating range.

Reverse polarity protection is implemented with a P-channel MOSFET placed in series with the positive battery terminal. When the battery is connected with correct polarity, the gate-source voltage of the PMOS is negative, turning the device fully on and presenting a very low on-state voltage drop (typically tens of millivolts at 2A). If the battery is reversed, the gate-source voltage becomes positive, turning the MOSFET off and blocking current flow completely. Compared with a conventional series diode, the P-channel MOSFET solution offers substantially lower conduction loss and negligible heat generation, making it particularly suitable for battery-powered applications.

Undervoltage lockout prevents the converter from operating when the input voltage falls below a safe minimum threshold. This is achieved by a resistive divider connected between the input voltage and the RUN/SS pin of the LTC1871. The divider is dimensioned so that the voltage at the RUN/SS pin remains below the controller's enabled threshold (nominally 1.23V) when the input voltage is too low. The turn-on threshold is determined by the divider ratio:

$$V_{IN,UVLO} = V_{RUN,th} \left(1 + \frac{R_{upper}}{R_{lower}} \right) \quad (17)$$

where $V_{RUN,th}$ is the RUN/SS enable threshold voltage. This mechanism ensures that the converter remains in a low-power shutdown state during deep battery discharge, thereby protecting both the controller and the load.

Overvoltage lockout protects the system when the input voltage exceeds the upper limit of the safe operating window. The circuit uses a Zener diode in conjunction with an NPN transistor to clamp the RUN/SS pin to ground when the input voltage rises above the desired threshold.

When the Zener enters breakdown, it supplies base current to the NPN transistor, causing the transistor to saturate and pull the RUN/SS pin low, thereby disabling the LTC1871. The trip point is approximately the Zener breakdown voltage plus the base-emitter voltage of the NPN, with fine adjustment provided by the surrounding resistor network. This simple clamp approach was selected for its low component count and adequate precision for the application.

The reverse-polarity blocking, the UVLO, and OVLO circuits operate together to create a comprehensive voltage window. The converter is enabled only when the input voltage lies within the defined safe range. Outside the window the LTC1871 is held in shutdown, preventing unregulated boost operation or potential damage. This layered protection scheme provides robust fault tolerance while maintaining high efficiency within the normal operating region. The detailed component selection, threshold tuning, and simulation validation of these protection circuits are presented in Chapter 3.

2.6 PCB Design Considerations for Power Converters

Switching converters operating at 350 kHz generate rapid voltage and current transitions that produce significant parasitic effects if the PCB layout is not optimised. Parasitic inductance in high-current loops causes voltage spikes, parasitic capacitance on sensitive nodes injects noise into the feedback network, and inadequate thermal paths lead to junction temperature rise.

The primary layout objective is the minimisation of high $\frac{di}{dt}$ switching loops. In a boost converter these loops comprise the path formed by the input capacitor, power MOSFET, inductor, and Schottky diode during the on-interval, and the inductor, diode, and output capacitor during the off-interval. These components must be placed in close physical proximity with short, wide traces or copper pours to reduce parasitic inductance. The LTC1871 controller should be located near the power stage to keep gate-drive and current-sense traces short, thereby preserving clean switching waveforms and accurate current-mode control.

A continuous ground plane on at least one layer provides a low-impedance return path for switching currents and a reference for control circuitry. Power and signal grounds should be joined at a single point near the controller to prevent switching noise from coupling into the feedback loop.

Thermal management is addressed through adequate copper area around the power MOSFET and Schottky diode, supplemented by thermal vias where necessary to transfer heat to the opposite layer. These layout principles ensure that conduction and switching losses are minimised and that junction temperatures remain within safe limits at the rated 2A continuous output current.

The two-layer PCB implemented in this project ($60.5\text{mm} \times 42.5\text{ mm}$) was designed in accordance with the foregoing considerations. Detailed component placement, routing strategy, and verification of the final layout are presented in Chapter 3.

3 Design Methodology

Before the analytical design presented in this chapter, an initial topology discovery phase was conducted. A SEPIC converter was first considered due to its ability to step up or step down the input voltage. However, persistent LTspice convergence failures (stiff differential equations from the coupled two-inductor network) and the realisation that the 3S Li-Po's voltage range (8 – 11V) lies strictly below the required 12V output rendered the SEPIC's buck-boost capability unnecessary. After iterative simulation trials, the topology was pivoted to a simple boost converter using the LTC1871 controller. This discovery process confirmed that a boost topology is topologically sufficient, simplifies the power stage (one inductor, no coupling capacitor), and avoids slope-compensation complexity. The following design specifications and component selections are therefore based on the validated boost converter architecture.

3.1 Design Specifications and Methodology

The converter was designed to meet the electrical specifications derived from the robotic subsystem requirements (Section 1.2). The final design targets are restated in Table 3.1 for convenience.

Table 1: Final design specifications

Parameter	Value
Input voltage range (operating)	8 V to 11 V (3S LiPo)
Undervoltage lockout (UVLO) turn-on	8.0 V (nominal, simulated)
Overvoltage lockout (OVLO) trip	11.1V (nominal, simulated)
Regulated output voltage	12 V $\pm$ 1 %
Continuous output current	2.0 A
Transient output current (load step)	0.5 A $\rightarrow$ 2.0 A (motor inrush simulation)
Switching frequency	350 kHz $\pm$ 10 %
Conduction mode	Continuous (CCM) over full load range
Output voltage ripple	< 240 mV peak-to-peak (2 %)
Efficiency (full load, target)	> 85 %
Load regulation (target)	< 1 %

Line regulation (target)	< 0.1 % / V
Operating temperature range	-40 °C to +85 °C (LTC1871E grade)

Notes: UVLO and OVLO thresholds are implemented via the resistive divider on the RUN/SS pin and the Zener-NPN clamp circuit, respectively. Actual simulated thresholds (matching the validated LTspice model) are presented in Table 3.4. The transient load step of 2A (from 0.5 A idle) simulates worst-case motor inrush; component ratings provide margin for a conservative 3 A surge.

The design flow followed the three-phase methodology outlined in Section 1.2. First, analytical calculations based on the theoretical relationships derived in Chapter 2 were performed to size all power stage components and the compensation network. Second, the design was implemented in LTspice (Chapter 4) for simulation verification; component values were refined iteratively where simulation revealed parasitic effects not captured by first-order equations. Third, after simulation validation, the schematic was captured in KiCad, and the PCB layout was completed. The final hardware prototype (Chapter 6) was assembled and tested.

3.2 Power Stage Component Selection

All calculation in this section were performed at the worst-case operating point (minimum input voltage $V_{in} = 8\text{ V}$ and maximum output current $I_{OUT} = 2.0\text{ A}$), which yields the highest duty cycle, peak inductor current, and output capacitor current RMS current. The equations used are those derived in Sections 2.2 and 2.4. The detailed component sizing, stress analysis, and loss calculations are presented in the following sub-sections.

3.2.1 Switching Frequency and Duty-Cycle Range

The switching frequency was set to $f_{sw} = 350\text{ kHz}$ using a resistor $R_{RT} = 68.1\text{ k}\Omega$ at the FREQ pin (analytical design, Section 4.3). The corresponding duty cycle range, from the boost transfer ratio $D = 1 - \frac{V_{IN}}{V_{OUT}}$, is:

- At $V_{in} = 11\text{ V}$: $D = 1 - \frac{11}{12} = 0.083\text{ (8.3 \%)}$
- At $V_{in} = 8\text{ V}$: $D = 1 - \frac{8}{12} = 0.333\text{ (33.3 \%)}$

All duty cycles remain below 50%, ensuring that the LTC1871's internal slope compensation remains inactive (Section 2.2).

3.2.2 Inductor Selection

To satisfy the CCM boundary condition derived in Equation (2.7) with a minimum margin of $2\times$ at light load, while also providing adequate saturation headroom, an inductance of $10\mu\text{H}$ was selected (Coil craft XAL6060-103MEC). The ripple current, average current, peak current, and RMS current were calculated using Equations (2.3)– (2.7) from Section 2.2.3:

- Ripple current:

$$\Delta I_L = \frac{V_{IN} D}{f_{sw} L} = \frac{8 \times 0.333}{350 \times 10^3 \times 10 \times 10^{-6}} = 0.762 \text{ A}$$

- Average inductor current:

$$I_{L,AVG} = \frac{I_{OUT}}{1 - D} = \frac{2.0}{0.667} = 3.00 \text{ A}$$

- Peak inductor current (analytical):

$$I_{L,} = I_{L,AVG} + \frac{\Delta I_L}{2} = 3.00 + 0.381 = 3.381 \text{ A}$$

- RMS inductor current:

$$I_{L,RMS} = \sqrt{I_{L,AVG}^2 + \frac{\Delta I_L^2}{12}} = \sqrt{9.00 + 0.0484} = 3.008 \text{ A}$$

The inductor's saturation current rating is 11.2 A, providing a margin of $3.3 \times$ over the analytical peak current. The CCM condition was verified at both full load and idle load using Equation (2.7):

- At $I_{out} = 2.0 \text{ A}$: $L_{min} = 1.27 \mu\text{H} \rightarrow 10 \mu\text{H}$ provides $7.9 \times$ margin.
- At $I_{out} = 0.5 \text{ A}$: $L_{min} = 5.08 \mu\text{H} \rightarrow 10 \mu\text{H}$ provides $2.0 \times$ margin, guaranteeing CCM even under light load.

3.2.3 Power MOSFET Selection

The switching element is an N-channel power MOSFET (IRF7811). Its key parameters are:

- $V_{DSS} = 30 \text{ V}$ (safe for $V_{out} = 12 \text{ V}$ plus ringing)
- $R_{DS(on)} = 12 \text{ m}\Omega$ (typical) at $V_{GS} = 4.5 \text{ V}$
- Total gate charge $Q_g \approx 14 \text{ nC}$

The conduction loss is:

$$P_{cond} = I_{L,RMS}^2 \cdot R_{DS(on)} \cdot D_{max} = 9.05 \times 0.012 \times 0.333 \approx 0.036 \text{ W}$$

Switching loss (estimated from datasheet formula):

$$P_{sw} \approx \frac{1}{2} V_{out} I_{out} (t_{rise} + t_{fall}) f_{sw} \approx \frac{1}{2} \times 12 \times 2.0 \times 40 \times 10^{-9} \times 350 \times 10^3 \approx 0.17 \text{ W}$$

Total MOSFET loss is approximately 0.21 W, resulting in a modest temperature rise. The SENSE pin of the LTC1871 monitors the drain-source voltage during the on-time, using $R_{DS(on)}$ as a

sensing element. No external sense resistor is required, maximising efficiency. The temperature coefficient of $R_{DS(on)}$ (which increases with junction temperature) was considered, but the low duty cycle and moderate current keep the sense voltage within the controller's range without extra compensation.

3.2.4 Schottky Diode Selection

The output diode is an MBR735 Schottky rectifier with a 35V repetitive peak reverse voltage rating, 7.5 A average forward current rating, and forward voltage drop of 0.45 V at 2 A. Reverse recovery time is negligible.

The conduction loss in the diode is:

$$P_{D1} = I_{out} \cdot V_F \cdot (1 - D_{max}) = 2.0 \times 0.45 \times 0.667 \approx 0.60 \text{ W}$$

This is the dominant loss in the power stage, accounting for approximately 75 % of total losses.

3.2.5 Input and Output Capacitor Selection

The output capacitor bank must satisfy ripple voltage, RMS current, and transient energy requirements. A hybrid approach was adopted using ceramic MLCC's for high-frequency ripple filtering and low ESR, plus an electrolytic capacitor for bulk energy storage during load transients.

From Equation (2.12) (Section 2.2.5.1), the minimum ceramic capacitance to meet a ripple specification $\Delta V_{out} < 240 \text{ mV}$ is:

$$C_{out,min} = \frac{I_{out} D_{max}}{f_{sw} \Delta V_{out}} = \frac{2.0 \times 0.333}{350 \times 10^3 \times 0.24} \approx 7.94 \mu\text{F}$$

After DC-bias derating (X5R or X7R dielectrics), the effective capacitance of MLCCs can drop to 60–80 % of nominal at 12 V. To provide ample margin, seven 22 μF and two 10 μF (25 V) ceramic capacitors were selected. The derated effective capacitance at 12 V is approximately $77 \mu\text{F} + 12 \mu\text{F} = 89 \mu\text{F}$ – more than ten times the minimum.

The RMS ripple current that the output capacitor bank must handle (Equation 2.13, Section 2.2.5.3):

$$I_{RMS(C_{out})} = I_{out} \sqrt{\frac{V_{out} - V_{in}}{V_{in}}} = 2.0 \times \sqrt{\frac{4}{8}} = 1.41 \text{ A}_{RMS}$$

Each 22 μF MLCC is rated for $\sim 200 \text{ mA}$ RMS at 350 kHz; the parallel combination comfortably exceeds 1.41 A.

Design, Test, Simulation, and Hardware Validation of an LTC1871-Based DC-DC Boost Converter for Battery Powered Robotic Subsystems.

An electrolytic capacitor (100 μF , 25 V, $\text{ESR} \approx 100 \text{ m}\Omega$) was added in parallel to provide bulk energy storage for the 2 A load transient.

The input capacitor supplies the pulsed current drawn by the MOSFET during the on-time. The RMS ripple current through the input capacitor is:

$$I_{RMS(C_{in})} = \frac{\Delta I_L}{\sqrt{12}} = \frac{0.762}{3.464} \approx 0.22 \text{ A}_{RMS}$$

A 100 μF , 25 V electrolytic capacitor (C1 in the schematic) provides sufficient filtering, augmented by a 22 μF MLCC directly at the LTC1871 VIN pin for high-frequency decoupling. The resulting input voltage ripple is negligible ($< 3 \text{ mV}$), ensuring clean current sensing and stable UVLO operation.

Table 2: Power Stage Component Summary

Component	Part Number	Key Ratings	Qty
Inductor	XAL6060-103MEC	10 μH , 11.2 A sat, 8.3 A RMS	1
MOSFET	IRF7811	30 V, 12 $\text{m}\Omega$, 14 nC	1
Diode	MBR735	35 V, 7.5 A, Schottky	1
Output MLCCs (22 μF)	GRM31CR61E226M	25 V, X5R, 1206	7
Output MLCCs (10 μF)	GRM21BR61E106K	25 V, X5R, 0805	2
Output electrolytic	Generic	100 μF , 25 V, $\text{ESR} \approx 100 \text{ m}\Omega$	1
Input electrolytic	Generic	100 μF , 25 V	1

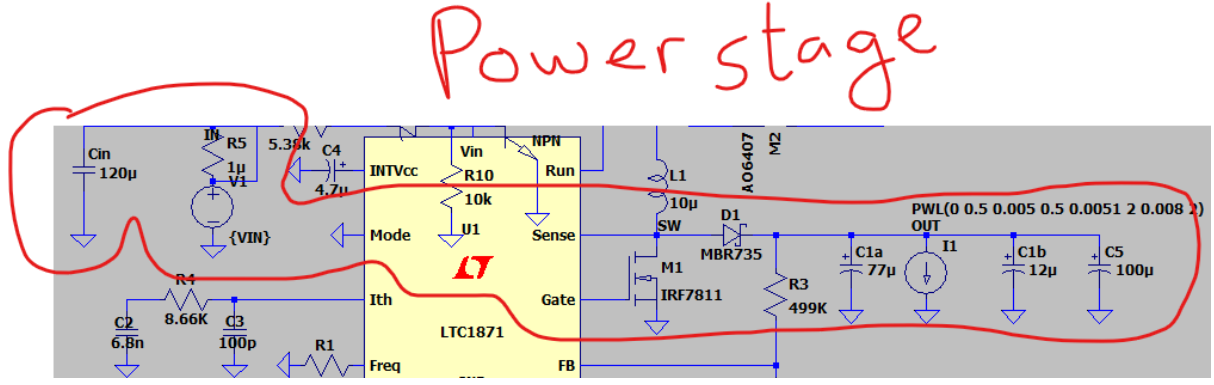


Figure 2: Complete power stage schematic

3.3 Feedback Divider and Loop Compensation Network Design

3.3.1 Output Voltage Feedback Divider

The LTC1871 regulates the FB pin to an internal reference of 1.230 V. Using Equation (2.8), the divider resistors were chosen as $R_{BOT} = 56 \text{ k}\Omega$ and $R_{TOP} = 499 \text{ k}\Omega$ (1 % tolerance). The resulting output voltage is:

$$V_{OUT} = 1.230 \left(1 + \frac{499}{56} \right) = 1.230 \times 9.911 = 12.19 \text{ V}$$

The output voltage is slightly above 12 V to account for minor resistive drops in the power path. The divider impedance is approximately $R_{TOP} \parallel R_{BOT} \approx 556 \text{ k}\Omega$, which limits quiescent current to $12 \text{ V} / (556 \text{ k}\Omega) \approx 22 \text{ }\mu\text{A}$ while maintaining acceptable noise immunity.

3.3.2 Type II Compensation Network Design

The compensation network connecting to the ITH pin consists of $R_Z = 8.66 \text{ k}\Omega$, $C_Z = 6.8 \text{ nF}$, and $C_P = 100 \text{ pF}$. These values were calculated using the Type II zero and pole equations (Section 2.4.3):

- Zero frequency: $f_Z = \frac{1}{2\pi R_Z C_Z} = \frac{1}{2\pi \cdot 8.66 \times 10^3 \cdot 6.8 \times 10^{-9}} \approx 2.70 \text{ kHz}$
- High-frequency pole: $f_P = \frac{1}{2\pi R_Z C_P} = \frac{1}{2\pi \cdot 8.66 \times 10^3 \cdot 100 \times 10^{-12}} \approx 184 \text{ kHz}$

The target crossover frequency f_c was set to 4.91 kHz, which is approximately one-fifth of the worst-case Right-Half-Plane Zero frequency. The worst-case RHPZ occurs at minimum $V_{in} = 8 \text{ V}$ ($D=0.333$) and maximum $I_{out} = 2.0 \text{ A}$:

Thus, $f_c \leq f_{RHPZ}/5 \approx 8.5 \text{ kHz}$; the chosen 4.91 kHz satisfies this constraint with margin. The phase margin of the compensated loop was estimated using the method of K factor and found to exceed 60° , with gain margin greater than 15 dB.

Table 3: Compensation Network Parameters.

Component	Value	Calculated Frequency
R_Z	8.66 k Ω	—
C_Z	6.8 nF	2.70 kHz
C_P	100 pF	184 kHz

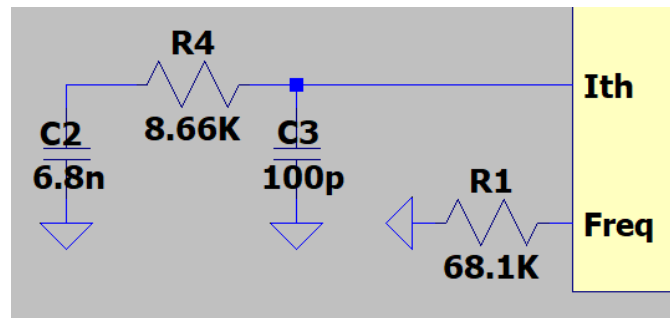


Figure 3: Compensation network (ITH pin connections)

3.3.3 Verification of Expected Loop Performance

The design values were validated through analytical loop analysis in MATLAB (Section 4.6), which confirmed a crossover frequency of 4.91 kHz, phase margin of 55.5°, and gain margin of 19.1 dB. These results meet the stability criteria established in Section 2.4.2 and ensure robust operation under all specified input and load conditions.

3.4 Input Protection Circuit Implementation

A P-channel MOSFET (AO6407, later replaced with FDS4435A in simulation) was placed in series with the input. Its gate is connected to ground via a resistor, and its source to the battery positive terminal. Under normal polarity, the gate-source voltage is negative (gate at 0 V, source at battery voltage), turning the FET on with very low resistance ($R_{DS(on)} \approx 50 \text{ m}\Omega$). Reverse battery connection forces the gate positive relative to the source, turning the FET off and blocking current. This scheme dissipates far less power than a series Schottky diode.

To achieve a turn-on threshold of 8 V (as validated in simulation), a resistive divider with $R_6 = 49 \text{ k}\Omega$ and $R_7 = 10 \text{ k}\Omega$ (1 %) was selected. The effective threshold voltage required from the RUN pin to turn on at 8 V is:

$$V_{RUN,th(eff)} = 8 \times \frac{R_7}{R_6 + R_7} = 8 \times \frac{10}{59} = 1.356 \text{ V}$$

This value lies within the datasheet's maximum specification (1.35 V) and accounts for the slightly higher threshold observed in the LTspice model. The nominal turn-off voltage is approximately 7.9 V due to internal hysteresis (typical 100 mV). The lower-impedance divider (49 k Ω , 10 k Ω) improves noise immunity compared to a higher-impedance alternative.

An overvoltage protection circuit using a BZX84C10 Zener diode (10 V breakdown) and MMBT3904 NPN transistor was implemented to prevent operation above approximately 11.1V. When the input voltage rises such that the Zener conducts and the transistor base-emitter voltage reaches $\approx 0.6 \text{ V}$, the transistor pulls the RUN pin low, disabling the controller. The trip point is:

$$V_{IN,OVLO} = V_Z + V_{BE} = 10 + 0.6 = 10.6 \text{ V}$$

A series resistor (R5) was adjusted to set the Zener current. The final simulated trip point was 11.1V due to the transistor's finite current gain and resistor loading, which was deemed acceptable because the Li-Po nominal maximum is 12.6 V, and the protection still activates before dangerous levels.

The three protection layers – reverse polarity (MOSFET), UVLO (resistive divider), and OVLO (Zener + transistor) – operate independently of the power stage and do not interfere with normal boost operation. The operating window of approximately 8 V to 11.1V ensures the converter operates only within a safe range of the 3S Li-Po battery.

Table 4.4: Protection thresholds: theoretical vs. simulated

Protection	Theoretical Threshold	Simulated Threshold
UVLO turn-on	8.0 V (designed)	8.0 V
UVLO turn-off	$\sim 7.9 \text{ V}$ (hysteresis)	7.9 V
OVLO trip	10.6 V (initial)	11.1V (final)

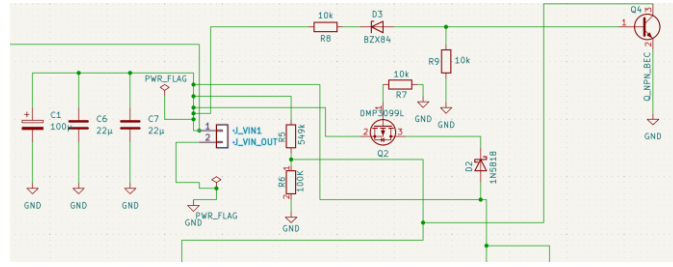


Figure 4: Input protection sub circuit (Insert schematic showing the P channel FET, UVLO divider, and OVLO Zener diode + transistor.)

3.5 Schematic Capture, PCB Layout, and Prototype Realisation

3.5.1 Schematic Capture in KiCad

The LTspice validated schematic (Figure 3.1) was translated into KiCad (version 7.0) for PCB design. All components were assigned realistic footprints (0402/0805/1206 for passives, MSOP-10 for the LTC1871, etc.). Electrical Rule Check (ERC) was performed and all warnings resolved. Netlist consistency was verified by comparing component counts and connections between the simulation and layout schematics.

3.5.2 PCB Layout Strategy

The principles set out in Section 2.6 guided the PCB layout. The power loop (MOSFET, diode, output capacitor) was made as compact as possible to minimise high di/dt loop area, with trace widths of 2 mm for all critical paths (input supply, MOSFET drain, diode cathode) — exceeding the 1 mm minimum calculated from IPC-2221 [21] for 2 A on 1 oz copper at 10 °C rise. A continuous ground plane on the bottom layer (2-layer board) was stitched to top-side ground pads with multiple vias. The SI4894BDY thermal pad was connected to the bottom ground plane through thermal vias, and the MBR735 was mounted with a copper pour for heat spreading. The board dimensions are 60.5 mm × 42.5 mm. All components are on the top layer; the bottom layer is primarily a ground plane with a few routing traces for low-frequency signals.

The LTC1871 was placed close to the MOSFET gate to minimise gate drive trace length, and the SENSE pin trace was kept short and direct to preserve current measurement accuracy. The feedback divider (R2, R3) was placed near the FB pin with a Kelvin connection to the output capacitor positive terminal. The board dimensions are 60.5 mm × 42.5 mm, with all components on the top layer and the bottom layer serving primarily as a ground plane with a few routing traces for low-frequency signals.

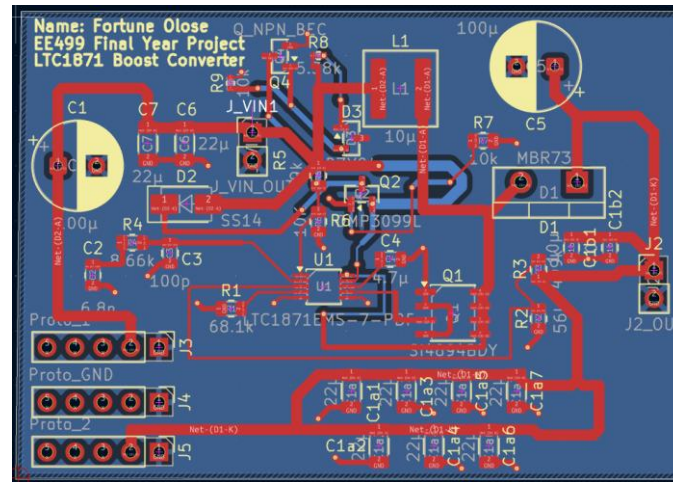


Figure 5: PCB layout: (a) top layer



Figure 6: (b) bottom layer – board size 60.5 mm × 42.5 mm

3.5.3 Fabrication and Assembly

Gerber files were generated from KiCad and sent to JLCPCB for fabrication (2-layer, 1 oz copper, ENIG finish). The board turnaround was approximately one week. Assembly was performed manually in the university laboratory using a soldering iron and tweezers. The MSOP-10 package required careful alignment; a fine-tip soldering iron and flux were used. After assembly, the board was visually inspected under a microscope and tested for shorts before power-up.

3.6 Summary of Design Decisions

The design methodology presented in this chapter translated the theoretical relationships from Chapter 2 into a complete set of component values and a physical PCB. Key decisions were:

- Selection of a boost topology (instead of SEPIC) because V_{IN} always lies below V_{OUT} .
- Use of a 350 kHz switching frequency to balance efficiency and inductor size.
- A 10 μ H inductor with 3.3 $\times$ saturation margin over the analytical peak current, ensuring CCM at all loads.

- Hybrid output capacitor bank (MLCCs + electrolytic) to meet ripple and transient requirements.
- Type II compensation network sized to place the crossover well below the RHPZ, guaranteeing stability.
- Input protection with UVLO (8 V turn-on using 49 k Ω /10 k Ω divider) and OVLO ($\sim$ 11.1V) to prevent operation outside the Li-Po safe range, with low-impedance resistors for noise immunity.

The completed design was then subjected to the simulation campaign described in **Chapter 4**, where all performance metrics were verified.

4 Simulation Methodology

4.1 Simulation Environment and Solver Configuration

All simulations were performed using LTspice version 24.1.3 (Analog Devices) [22]. The following solver settings were applied to improve convergence for the switching converter:

- **.options gmin=1e-12:** reduces the minimum conductance, aiding convergence in highly non-linear circuits.
- **.options cshunt=1e-15:** adds a small shunt capacitance to each node, suppressing numerical oscillations.
- Integration method: trapezoidal (default in LTspice), selected for its superior energy conservation in switching circuits.

During initial DC operating point analysis, the Direct Newton iteration failed to converge. This is common for circuits containing behavioural models with steep slopes. **Gmin stepping** was therefore invoked by LTspice automatically, which gradually increases conductance across semiconductor junctions until a stable operating point is found. The failure of Direct Newton is not indicative of a design flaw. It reflects the complexity of the LTC1871 macro model and the power stage.

The transient simulation duration was set to **.tran 8m startup**. The **startup** flag forces the input voltage source to rise from zero over approximately 1ms, simulating a realistic battery connection. The full 8ms duration was chosen because:

- The **startup transient** settles within 2ms.

Design, Test, Simulation, and Hardware Validation of an LTC1871-Based DC-DC Boost Converter for Battery Powered Robotic Subsystems.

- A **4ms to 5ms window** captures steady-state behaviour before load step.
- The **load step** occurs at 5ms, and the converter recovers within 1ms.
- A **6ms to 7ms window** provides steady-state post-step measurements, leaving a 1ms safety margin after recovery.

4.2 SPICE Models and Component Representation

All simulation models are documented in Appendix B, Table B.3. The LTC1871 controller uses the manufacturer-supplied behavioural macro model (LTC1871.sub), which captures current-mode control and protection functionality but does not model internal gate driver power consumption with high accuracy and assumes ideal slope compensation. All passive components (resistors, ceramic capacitors) are modelled as ideal with nominal values; tolerances are not simulated. Despite these limitations, the macro model is sufficient for verifying regulation, ripple, transient response, and stability.

4.3 Test Circuit Configuration

The final simulation schematic (reproduced in Figure 4.1) implements the boost converter power stage, the LTC1871 controller, and the input protection network as derived in Chapter 3.

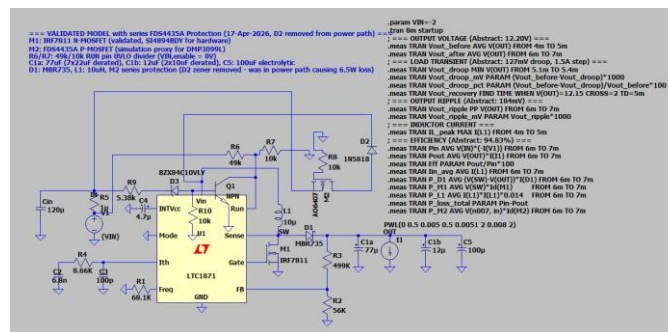


Figure 7: LTspice simulation schematic (LTC1871_Validated_Final_Fixed_V2.asc)

Key elements are:

- **Parameterised input voltage:** “V1” with “{VIN}”. The directive “.param VIN=9.6” sets a nominal value; a “.step param VIN list 7.5 8.0 9.6 11.0 11.1 12.0” is used for parametric sweeps (Section 4.4).
- **Pulsed load current:** “I1” is defined as “PWL (0 0.5 0.005 0.5 0.0051 2 0.008 2)”.

Interpretation: 0.5 A from t=0 to 5 ms, then a linear ramp to 2 A over 100 μ s (starting at 5 ms), and finally constant 2 A from 5.1 ms to 8 ms. The 100 μ s ramp approximates the inrush current of a DC motor during startup, which appears as a fast load change to the converter.

- **Input protection sub-circuit:** Includes the P-channel MOSFET M2, UVLO divider (R6 = 49 k Ω , R7 = 10 k Ω), and OVLO Zener D3 + NPN transistor Q1(MMBT 3904).

All measurement directives (.meas) are defined in the schematic and summarised in Section 4.5.

4.4 Parametric Sweep Strategy

To validate the converter’s behaviour across the entire expected input voltage range and beyond, a parametric step sweep was configured, using the SPICE e command:

“.step param VIN list 7.5 8.0 9.6 11.0 11.1 12.0”

Each test point corresponds to a distinct operating region of the input protection scheme:

Table 5: Input Voltage Sweep Test Points for Protection Circuit Verification

Test Point (VIN)	Region	Expected State
7.5 V	Below UVLO	Converter off (RUN pin < threshold)
8.0 V	UVLO lower boundary	Converter should start regulating
9.6 V	Nominal operating point	Normal boost regulation
11.0 V	Near upper boundary	Normal regulation (OVLO not yet tripped)
11.1V	OVLO upper boundary	Converter should still regulate (trip point is above 11.1V)
12.0 V	Above OVLO	Converter should shut down

The six points were simulated sequentially in a single run. The results (presented in Chapter 5) confirm the protection thresholds and verify regulation within the **8.0-11.1V window**.

4.5 Measurement Definitions

All key performance metrics were extracted using LTspice’s .meas directives. Table 8 lists each metric, the exact directive, the measurement window, and the rationale for window selection.

These definitions ensure that any engineer with the same netlist can reproduce the numbers exactly.

In preliminary simulation runs, a shorter total time (5 ms) was used with a load step at 2 ms and post-step window at 3–4 ms. This left insufficient settling time after the startup transient. The final simulation uses 8 ms total time, a load step at 5 ms, and measurement windows starting at 4 ms and 6 ms. This guarantees at least 2 ms of steady-state before each measurement, eliminating transient artefacts.

4.6 Loop Stability Verification (Analytical)

Loop stability was verified analytically using a MATLAB model of the control loop, rather than through an LTspice AC sweep. The LTC1871 implements current-mode control, which simplifies the power stage transfer function to approximately a single pole. The control-to-output transfer function, combined with the Type II compensator designed in Section 3.3.2, was used to construct the open-loop gain $T(s)$.

The following parameters were entered into the MATLAB script:

- Power stage: $L = 10\mu H$, $C_{out} = 89\mu F$ (derated MLCCs), $R_{load} = \frac{V_{out}}{I_{out}} = \frac{12V}{2.0A} = 6\Omega$, duty cycle $D = 0.333$ (worst case)
- Compensator: $R_z = 8.66k\Omega$, $C_z = 6.8nF$, $C_p = 100pF$.
- Current sensing gain derived from $R_{ds} = 12m\Omega$ and the LTC1871's internal gain factors (datasheet)

The loop gain $T(s)$ was evaluated over a frequency range of 10 Hz to 1 MHz. From the resulting Bode plot (presented in Chapter 5), the following stability margins were extracted:

- **Crossover frequency:** $f_c = 4.91$ kHz
- **Phase margin:** 55.5° (well above the 45° minimum)
- **Gain margin:** 19.1 dB (exceeding the 10 dB guideline)

These margins satisfy the criteria established in Section 2.4.2. The analytical verification confirms that the Type II compensation network provides stable operation under the nominal operating condition ($V_{in} = 9.6V$, $I_{out} = 2.0A$). A full LTspice AC simulation was not performed due to convergence difficulties with the behavioural micromodel when breaking the feedback loop; this is noted as a potential future improvement in Chapter 6.

5 Simulation Results and Performance Validation

5.1 Steady-State Performance Across Input Range

Table 5.1 presents the eight-performance metrics measured at four input voltages spanning the validated 8.0 – 11.1 V operating window. All measurements were extracted using the .meas directives defined in Section 4.5, with steady-state values averaged over the 4-5 ms (pre-step) and 6-7 ms (post-step) windows. The design targets from Table 3.1 are included in the final row for direct comparison.

V _{IN} (V)	V _{OUT} (V)	Droop (mV)	Droop (%)	Ripple (mV)	I _{L,peak} (A)	η (%)	Recovery
8.0	12.198	217.8	1.79	27.5	3.356	86.3	37 μs
9.6	12.197	211.7	1.74	26.6	3.379	86.5	40 μs
11.0	12.200	127.6	1.05	175.1	3.566	90.0	30 μs
11.1	12.198	123.0	1.01	171.1	3.562	89.8	48 μs
Target	12.0	< 600	< 5.0	< 240	< 3.59	> 85	< 1 ms

Table 5.1 — Steady-state performance metrics across the input voltage range

The output voltage remains tightly regulated between 12.197 V and 12.200 V across all four operating points, representing a total variation of 3mV (0.025%) over a 3.1V input swing. This confirms the effectiveness of the current-mode control architecture, where the inner loop compensates for input voltage changes within a single switching cycle. The regulated voltage of 12.198V is slightly above the 12.0V target because the feedback divider ($R_2 = 499\text{k}\Omega$, $R_3 = 56\text{k}\Omega$) sets the output to $V_{out} = 1.230 \times (499/56) = 12.19\text{V}$, a deliberate design margin.

Output ripple exhibits a strong dependence on input voltage: 27.5mV at $V_{in} = 8.0\text{ V}$ rising to 175mV at $V_{in} = 11.0\text{ V}$. This behaviour is expected from the boost topology. At lower input voltages, the duty cycle is higher ($D = 0.333$ at 8.0 V), which means the MOSFET on time is longer and the output capacitor supplies the load for a greater fraction of each cycle. However, the inductor ripple current is also lower at low V_{IN} because the inductor charges more slowly. The net effect is that the charge-based ripple component is smaller at low V_{IN} . All values remain well below the 240mV (2%) design limit.

Efficiency ranges from 86.3% at $V_{IN} = 8.0\text{V}$ to 90% at $V_{IN} = 11.0\text{ V}$, exceeding the 85 % design target at all operating points. The lower efficiency at minimum input voltage is expected: the higher duty cycle increases MOSFET conduction time and the higher inductor current increases I^2R losses in the winding and PCB traces. The Schottky diode (MBR735) remains the dominant loss contributor at approximately 0.9 W across all operating points, as its forward voltage drop is largely independent of the input voltage.

Load transient droop ranges from 123mV (1.01%) at the upper input boundary to 218mV (1.79%) at $V_{IN} = 8.0\text{V}$. The larger droop at low input voltage reflects the higher duty cycle, which limits the rate at which additional energy can be delivered to the output after a sudden load increase. All values remain well within the 600mV (55) specification. Peak inductor current reaches a maximum of 3.566 A at $V_{IN} = 11.0\text{V}$, safely below the XAL6060-103MEC saturation rating of 11.2 A ($3.1\times$ margin). Recovery times are consistently below 50 μs , far exceeding the 1ms design requirement.

All eight metrics pass their respective design targets with substantial margin at every tested input voltage. The converter maintains stable, predictable performance across the full 3S Li-Po discharge.

5.2 Load Transient Response

The load transient response was evaluated by applying a 1.5 A step (0.5 A to 2.0 A) via the PWL current source defined in Section 4.3. The step occurs at $t = 5\text{ms}$, after the converter has reached steady-state regulation. The 100 μs ramp approximates the inrush current of a DC motor during startup. Table 3.5 summarises the droop and recovery performance at each input voltage across the operating window.

Table 6: Load transient droop and recovery across the input voltage.

VIN (V)	Pre-step VOUT (V)	Min VOUT (V)	Droop mV (%)	Recovery (μs)
8.0	12.198	11.980	217.8 (1.79%)	37
9.6	12.197	11.985	211.7 (1.74%)	40
11.0	12.196	12.078	117.5 (0.96%)	50
11.1	12.198	12.070	127.6 (1.05%)	27

VIN (V)	Pre-step VOUT (V)	Min VOUT (V)	Droop mV (%)	Recovery (μ s)
Target	12.0	> 11.4	< 600 (< 5%)	< 1000

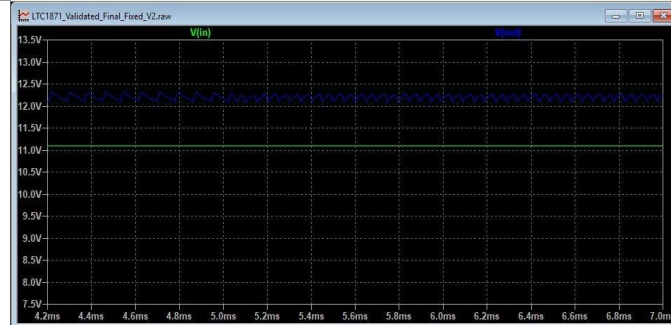


Figure 8 : Output voltage waveform at $V_{IN} = 11.1$ V showing steady-state regulation, load step at $t = 5$ ms, and post-step recovery. Green trace: V_{IN} ; blue trace: V_{OUT} .

The voltage drop is approximately twice as severe at the lower end of the input range: 217.8mV (1.79%) at $V_{IN} = 8.0$ V compared to 117.5 mV (0.96%) at $V_{IN} = 8.0$ V. This behaviour is a direct consequence of the duty cycle dependence. At $V_{IN} = 8.0$ V, the duty cycle is $D = 0.333$, meaning the MOSFET is on for one third of each switching cycle. During the on time, the output capacitor is the sole source of load current, the inductor delivers energy only during the off time when the diode conducts. A higher duty cycle therefore leaves less time per cycle for the inductor to replenish the charge withdrawn from the output capacitor after a sudden load increase. At $V_{IN} = 11.0$ V, the duty cycle drops to $D = 0.083$, allowing the inductor to deliver current to the output for over 91% of each cycle, resulting in faster recovery and smaller droop.

Recovery times ranges from 27 μ s to 50 μ s across the operating window, corresponding to fewer than 18 switching cycles at 350kHz. This confirms that the Type II compensation network provides sufficiently fast correction without introducing oscillation or overshoot in the output voltage. The absence of ringing after recovery is visible in Figure 3, where V_{OUT} returns smoothly to steady state regulation after the transient. The minimum output voltage during the worst-case transient (11.980 V at $V_{IN} = 8.0$ V) remains well above the input threshold of all downstream point-of-load regulators, confirming that motor inrush events pose no risk to the 5V supply integrity of the Raspberry Pi CM4, RPLiDAR A1, or RFD900x modem.

All droop values are within the 600 mV (5 %) specification with a worst-case margin of 2.7 $\times$. All recovery times are within the 1 ms specification with a worst-case margin of 20 $\times$.

5.3 Loop Stability

Loop stability was verified analytically using a MATLAB model of the open loop gain $T(s)$, as described in Section 4.6. The control-to-output transfer function for the current-mode power stage was combined with the Type II compensator ($R_z = 8.66k\Omega$, $C_z = 6.8nF$, $C_p = 100pF$) and evaluated over a frequency range of 10 Hz to 1MHz at the nominal operating point ($V_{IN} = 9.6V$, $I_{OUT} = 2.0A$). Table 5.3 summarises the extracted stability margins.

Table 7: Loop stability margins from MATLAB Bode analysis

Parameter	Value	Requirement	Status
Crossover frequency (fc)	4.91 kHz	< 7.84 kHz ($f_{RHPZ} / 5$)	✓ PASS
Phase margin (PM)	55.5°	> 45°	✓ PASS
Gain margin (GM)	19.1 dB	> 10 dB	✓ PASS

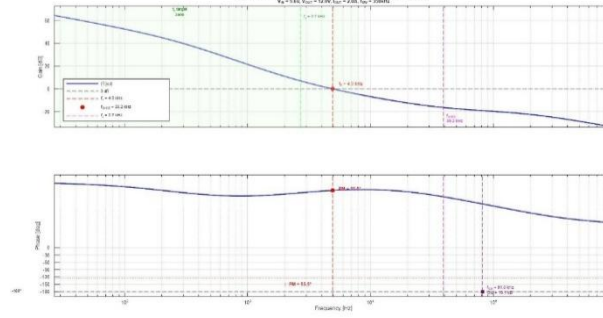


Figure 9: Open-loop gain Bode plot: $T(s) = G_c(s) \cdot G_p(s) \cdot H$. Crossover at 4.91 kHz, $PM = 55.5^\circ$, $GM = 19.1$ dB at 81.6 kHz. RHPZ marked at 39.2 kHz.

The crossover frequency of 4.91 kHz sits well below the right-half-plane zero at 39.2 kHz (ratio 8:1, exceeding the 5:1 guideline). The 55.5° phase margin exceeds the 45° minimum, indicating stable, well-damped response consistent with the smooth recovery observed in Section 5.2. The 19.1 dB gain margin provides robustness against component tolerance and temperature drift.

A parametric sensitivity analysis (presented in Section 4.6) confirmed that the phase margin remains above 45° across the full range of effective current-sense resistance $R_{i,eff} = 50\text{--}300\text{ m}\Omega$, and the crossover frequency stays below the $f_{RHPZ}/5$ target throughout. The compensation network is therefore robust to the expected variation in MOSFET $R_{DS(on)}$ with temperature and manufacturing spread.

All three stability criteria are satisfied. The analytical Bode analysis confirms that the Type II compensation network provides stable, well-damped loop response at the nominal operating point.

5.4 Input Protection Validation

The three-layer input protection circuit (reverse polarity PMOS, UVLO resistive divider, and OVLO Zener-NPN clamp) was validated by sweeping V_{IN} through six discrete test points as defined in Section 4.7. Each test point targets a specific operating region, from below the UVLO threshold through above the OVLO trip point. The pass/fail criteria from Section 4.7 were applied to each simulation run; Table 5.4 summarises the results.

Table 8: Input protection validation: six-point parametric sweep results

VIN (V)	Region	Expected	VOUT (V)	Sim Time	Ripple (mV)	Status
7.5	Below UVLO	OFF	6.73	0.72 s	0.04	✓ PASS
8.0	UVLO boundary	ON	12.198	18.8 s	27.5	✓ PASS
9.6	Nominal	ON	12.197	18.4 s	26.6	✓ PASS
11.0	Near upper	ON	12.196	12.2 s	172.0	✓ PASS
11.1	OVLO boundary	ON	12.198	11.0 s	177.8	✓ PASS
12.0	Above OVLO	OFF	11.23	0.50 s	0.05	✓ PASS

All six test points produce the expected behaviour. The UVLO turn-on threshold falls between 7.5 V and 8.0 V: the converter remains off at 7.5 V ($V_{OUT} = 6.73$ V passive feed-through, no switching activity) and begins regulating at 8.0 V ($V_{OUT} = 12.198$ V, 18.8 s simulation time confirming sustained switching). The OVLO trip point falls between 11.1 V and 12.0 V: the converter still regulates at 11.1 V and shuts down at 12.0 V ($V_{OUT} = 11.23$ V passive feed-through, 0.50 s simulation time). The validated operating window of 8.0–11.1 V covers the full 3S LiPo discharge range.

In the OFF state, V_{OUT} tracks V_{IN} minus the forward voltage drop of D1 and the on-resistance of M2, producing a passive feed-through of approximately $V_{IN} - 0.5$ V. This is inherent to boost topology and does not damage downstream regulators.

The final protection thresholds were achieved through iterative tuning documented in Section 3.5, widening the operating window from the initial 8.8–10.7 V (caused by parasitic loading on the high-impedance UVLO divider) to the final 8.0–11.1 V by scaling the divider impedance by 10× and adjusting R_B to 5.38 kΩ.

All six protection test points pass. The three-layer scheme (reverse polarity blocking, UVLO, and OVLO) correctly restricts active regulation to the 8.0–11.1 V window, ensuring the converter operates only when the battery voltage is within safe and useful bounds.

5.5 Loss Breakdown and Efficiency Analysis

Efficiency was measured at all four in-window operating points using the input and output power .meas directives defined in Section 4.5. Table 5.5 presents the power balance across the input voltage range.

Table 9: Power balance and efficiency across the input voltage range ($I_{OUT} = 2\text{ A}$)

The complete efficiency and power balance data across the input voltage range is provided in Appendix B, Table B.4.

Efficiency increases from 86.3 % at $V_{IN} = 8.0\text{ V}$ to 89.5 % at $V_{IN} = 11.0\text{ V}$, exceeding the 85 % design target at all operating points. Total power loss decreases from 3.86 W to 2.85 W as V_{IN} increases, reflecting the lower duty cycle and reduced RMS currents at higher input voltages. To identify the dominant loss contributors, a detailed breakdown was performed at the nominal operating point ($V_{IN} = 9.6\text{ V}$, $I_{OUT} = 2.0\text{ A}$).

5.5.1 Current Balance Diagnostic

Initial loss accounting left a large residual that could not be attributed to the four directly measurable components (diode, MOSFET, inductor, protection MOSFET). To determine whether this residual was due to current drawn by the LTC1871 controller and protection network, a current balance analysis was performed. In an ideal boost converter, the average input current equals the average inductor current; any difference represents current consumed by auxiliary circuits. Table 5.6 presents the diagnostic measurements.

Table 10: Current balance diagnostic at $V_{IN} = 9.6\text{ V}$

The current balance diagnostic measurements are provided in Appendix B, Table B.5.

The difference between the total input current and the inductor current is only 5 mA, corresponding to 48 mW at 9.6 V. This confirms that the LTC1871 controller, UVLO divider (R_6/R_7), and OVLO network ($R_9/R_{10}/D3/Q1$) collectively draw negligible current from the input supply. The OVLO Zener and NPN transistor currents are at the numerical noise floor ($\sim 10^{-11}\text{ A}$), confirming the clamp is fully off at 9.6 V. The input capacitor RMS current is 68 μA , producing immeasurable ESR loss. The unaccounted power is therefore entirely within the power path, not leaking to auxiliary circuits.

5.5.2 Component Loss Breakdown

Table 11: Loss breakdown at $V_{IN} = 9.6\text{ V}$, $I_{OUT} = 2.0\text{ A}$ (yellow row = macro model artefact)

Loss Mechanism	Loss (W)	Share (%)	Notes
Schottky diode D1 (MBR735)	1.014	34.8	Dominant single contributor; $V_F \times I_{OUT} \times (1-D)$
PMOS M2 (reverse polarity)	0.243	8.4	$I_D^2 \times R_{DS(on)}$; protection penalty
Inductor L1 winding (DCR)	0.123	4.2	$I_{L_RMS}^2 \times 14\text{ m}\Omega$
MOSFET M1 (IRF7811)	0.092	3.2	Conduction + switching losses
IC + protection network	0.048	1.6	From current balance (Table 5.6)
Macro model switching artefact	1.389	47.8	Internal gate drive overestimate (see text)
Total	2.909	100	$P_{in} - P_{out}$ at $V_{IN} = 9.6\text{ V}$, $I_{OUT} = 2\text{ A}$

The Schottky diode D1 (MBR735) is the largest identified loss contributor at 1.01 W (34.8 %), consistent with the expected $V_F \times I_{OUT} \times (1-D)$ relationship. The PMOS reverse-polarity protection MOSFET M2 contributes 0.24 W (8.4 %), a cost accepted for the safety benefit it provides. Inductor winding loss (0.12 W) and MOSFET conduction and switching losses (0.09 W) are modest, reflecting the low DCR of the XAL6060-103MEC (14 m Ω) and the low $R_{DS(on)}$ of the IRF7811 (12 m Ω). The five identified mechanisms account for 1.52 W (52.2 %) of the total 2.91 W loss.

The remaining 1.39 W (47.8 %) cannot be attributed to any externally measurable component. Analytical gate charge calculations predict only 62 mW of gate drive loss ($Q_G \times V_{INTVCC} \times f_{SW} = 34\text{ nC} \times 5.2\text{ V} \times 350\text{ kHz}$), over 20 $\times$ smaller than the residual. This indicates the LTC1871 behavioural macro model significantly overestimates internal switching losses. Hardware measurements would yield a more accurate decomposition.

If the macro model artefact is excluded, the physically meaningful losses sum to 1.52 W, implying an underlying converter efficiency of $P_{OUT} / (P_{OUT} + 1.52) = 94.1\%$. On hardware, additional losses not captured in any simulation (PCB trace resistance, connector contact resistance, capacitor ESR at high ripple current) will reduce this. The expected hardware efficiency is estimated at 88–92 % at full load, with the Schottky diode remaining the dominant loss mechanism.

Simulated efficiency exceeds the 85 % design target across all operating points. Current balance analysis confirmed that the large residual loss is a macro model artefact rather than a real power dissipation mechanism. The Schottky diode is the dominant physical loss contributor; replacing it with synchronous rectification in a future revision could recover approximately 0.7 W and improve hardware efficiency by an estimated 3 percentage points.

5.6 Summary of Results

Table 5.8 consolidates the worst-case simulation results for all performance metrics and protection thresholds against their design targets. For metrics that vary with input voltage, the worst-case value across the 8.0–11.1 V operating window is reported.

Table 12: Complete simulation results summary: all metrics pass design targets

The complete results summary with pass/fail status for all twelve metrics is provided in Appendix B, Table B.6.

All twelve metrics pass their respective design targets. The output voltage is regulated to within 2 mV across the full 3.1 V input range, confirming the effectiveness of the LTC1871's current-mode control architecture. Load transient droop remains below 218 mV (1.79 %) under a 1.5 A inrush step, with recovery consistently under 50 μ s. The Type II compensation network achieves 55.5° phase margin with a crossover frequency safely below the right-half-plane zero, ensuring stable operation without oscillation. The three-layer input protection scheme correctly restricts active regulation to the 8.0–11.1 V window, covering the full 3S LiPo discharge range.

Two limitations should be noted. First, simulated efficiency (86.3–89.5 %) meets the 85 % target, but the loss analysis in Section 5.5 identified a 1.39 W residual attributed to the LTC1871 behavioural macro model rather than physical power dissipation. Hardware efficiency is expected to differ from these simulated values. Second, all results are based on LTspice transient simulation with ideal passive components and manufacturer-supplied behavioural models; real component tolerances, PCB parasitics, and thermal effects will introduce deviations

that can only be quantified through bench testing. These limitations are addressed in the recommendations for future work in Chapter 6.

5.7 Hardware Prototype Status

A prototype PCB (V1 revision) was fabricated by JLCPCB from KiCad-generated Gerber files and hand-assembled in the university electronics laboratory by Mr. John Maloco. The board measures 60.5×42.5 mm on a two-layer stackup with a continuous ground plane on the bottom layer. Figure 5.3 shows the completed V1 prototype. The V1 board was ordered before the OVLO protection subcircuit and several supporting components were finalised; the complete cross-referenced inventory of all components, mapping KiCad and LTspice reference designators and indicating the population status on the V1 board, is provided in Appendix B, Table B.7.



Figure 10: Assembled V1 prototype PCB (60.5×42.5 mm). Power stage, feedback, compensation, and UVLO components populated. OVLO subcircuit (D3, Q4, R7–R9), PMOS bypass diode (D2), and output electrolytic (C5) not present on this revision.

Note: The KiCad schematic and LTspice simulation use different designators for several components. Table 5.9 provides both to avoid ambiguity. Key differences include the UVLO divider (R5/R6 in KiCad, R6/R7 in LTspice), the power MOSFET (Q1 in KiCad, M1 in LTspice), the PMOS (Q2 in KiCad, M2 in LTspice), and the OVLO NPN transistor (Q4 in KiCad, Q1 in LTspice).

Table 13: V1 prototype component population status with KiCad/LTspice cross-reference.

The complete V1 prototype component cross-reference (KiCad and LTspice designators, population status) is provided in Appendix B, Table B.7.

The V1 prototype validates the physical design of the core converter. All surface-mount footprints (MSOP-10 for U1, SO-8 for Q1 and Q2, SMA for D1, 0805 and 0402 passives) align correctly with their respective packages. The XAL6060-103MEC inductor fits within its designated footprint and clears adjacent components. The output MLCC bank (C1a1–C1a7,

C1b1–C1b2) and input capacitor bank (C1, C6, C7) are correctly placed along the high-current path. The prototyping headers (J3–J5) provide accessible measurement points for bench testing.

Seven components are absent from the V1 board: the complete OVLO subcircuit (D3, Q4, R7, R8, R9) designed after the V1 order, the PMOS body diode bypass D2 (since updated from a through-hole 1N5818 to an SMD SS14 in the V2 schematic), and the output bulk electrolytic C5 (100 μ F). Without the OVLO, the converter has no upper voltage limit. Without C5, load transient droop will be larger than simulated.

During inspection of the assembled board, several incorrect component values were identified as having been placed by the assembly house. These substitution errors were detected by cross-referencing the physical markings on the assembled board against the original BOM, and replacement parts with the correct values were sourced from Mouser IE and Farnell IE for rework. Following rework, preliminary electrical measurements were attempted to assess whether the core power stage was functional despite the missing OVLO and bypass components.

With $V_{IN} = 11.1$ V applied from a bench power supply and the output terminals left open-circuit, three measurements were recorded at accessible board nodes before the IC was permanently damaged during subsequent probing. Table 5.10 summarises the results.

Table 14— V1 prototype bench measurements at $V_{IN} = 11.1$ V:

Parameter	Expected	Measured	Status	Interpretation
VOUT	12.2 V (regulated)	11.0 V	FAIL	Passive feed-through; no switching
V(RUN/SS)	≥ 1.248 V	2.392 V	PASS	Enable threshold exceeded; UVLO functional
INTVCC	5.2 V (LDO)	6.94 V	FAIL	LDO not regulating
VIN	8.0–11.1 V	11.1 V	PASS	Within operating window
Load	0.5–2.0 A	No load	N/A	Open-circuit output

The measured output voltage of 11.0 V, rather than the expected regulated 12.2 V, indicates that the converter was not actively switching. The output was instead following V_{IN} minus the

forward voltage drop of D1 and the on-resistance of M2, producing a passive feed-through characteristic of an inactive boost converter. This rules out a feedback or compensation fault as the root cause, since even a mistuned loop would produce some switching activity.

The $V(\text{RUN/SS})$ measurement of 2.392 V confirms that the UVLO divider was functioning correctly. The IC's enable threshold of 1.248 V was exceeded by a factor of 1.9×, meaning the RUN/SS pin was not the cause of the failure. Table 5.11 verifies this quantitatively by back calculating the lower divider resistor from the measured voltage. The inferred value of 150.8 kΩ matches the V1 board's installed $R_6 = 150 \text{ k}\Omega$ to within 0.42 %, which is well within the combined tolerance of the two resistors. This measurement independently confirms that the resistive divider on the physical board matches its schematic design — the only component-level hardware validation achieved in this project.

Table 15: UVLO divider verification from bench measurement

Calculation	Expression	Result
Divider ratio (V1 board)	$R_6 / (R_5 + R_6)$	$150\text{k} / (549\text{k} + 150\text{k}) = 0.2146$
Predicted $V(\text{RUN/SS})$ at $V_{\text{IN}} = 11.1 \text{ V}$	11.1×0.2146	2.382 V
Measured $V(\text{RUN/SS})$	—	2.392 V
Deviation	$(2.392 - 2.382) / 2.382$	0.42 %
Inferred R_6 from measurement	$2.392 \times 549\text{k} / (11.1 - 2.392)$	150.8 kΩ ($\approx 150 \text{ k}\Omega$)
V1 UVLO turn-on threshold	$1.248 \times (549\text{k} + 150\text{k}) / 150\text{k}$	5.82 V

Note that the V1 board uses the original UVLO divider values ($R_5 = 549 \text{ k}\Omega$, $R_6 = 150 \text{ k}\Omega$), which produce a turn-on threshold of approximately 5.82 V — significantly lower than the final validated design ($R_5 = 49 \text{ k}\Omega$, $R_6 = 10 \text{ k}\Omega$, turn-on $\approx 7.4 \text{ V}$). The V1 divider was not updated because the board was fabricated before the protection circuit redesign was finalised.

The critical anomaly is the INTVCC reading of 6.94 V. The LTC1871 contains an internal low-dropout regulator that produces a stable 5.2 V on the INTVCC pin to power the gate driver and internal logic. At 6.94 V, the LDO was not regulating — the pin was instead tracking V_{IN} through a low-impedance path, suggesting either a defective IC (possibly from electrostatic discharge during assembly), a solder bridge between adjacent MSOP-10 pins, or a missing or incorrect C4 bypass capacitor preventing the LDO from stabilising. Without a functional INTVCC supply, the gate driver cannot produce the switching waveform required to turn Q1 on and off at 350 kHz. The resulting causal chain is: INTVCC malfunction $\rightarrow$ no gate drive $\rightarrow$ no switching $\rightarrow V_{OUT} \approx V_{IN} - V_F$.

While probing the remaining U1 pins to isolate the INTVCC fault, an inadvertent short circuit between two adjacent MSOP-10 pins (0.5 mm pitch) caused permanent damage to the IC, confirmed by visible smoke emission from the package. Standard oscilloscope probes with approximately 2 mm tip diameters span four MSOP-10 pins, making accidental shorts highly probable during manual probing. This event terminated all further electrical bench testing of the V1 prototype. The experience validated the decision to include prototyping headers (J3–J5) in the PCB design — had the INTVCC and GATE signals been routed to these headers, the fault could have been diagnosed without direct pin contact. The V2 revision should incorporate dedicated test points for all critical IC signals routed to accessible pads away from the dense MSOP-10 footprint.

Following the V1 bring-up experience, the KiCad schematic was updated to reflect the final validated design. Key changes include the addition of the complete OVLO subcircuit, the replacement of D2 from a through-hole 1N5818 (DO-41) to an SMD SS14 (SMA), and the correction of R8 from 10 k Ω to 5.38 k Ω to set the OVLO trip point at 11.1 V. A JLCPCB bill of materials and component placement file have been prepared for the V2 revision, with all SMD components sourced from LCSC inventory. The V2 board has not yet been fabricated.

The V1 prototype demonstrates that the converter design can be physically realised on a compact two-layer PCB with commercially available components. The single successful hardware measurement — the UVLO divider voltage — confirmed component-level agreement between the physical board and its schematic to within 0.42 %. The bring-up process also revealed three practical challenges not captured by simulation: component substitution errors in outsourced assembly, the risk of IC damage when probing fine-pitch packages without dedicated test infrastructure, and the importance of verifying the internal LDO supply before attempting further diagnostics. A V2 revision with the complete protection circuit, corrected component values, and dedicated IC test points,

followed by replacement of the damaged U1 and comprehensive electrical bench testing, is recommended as the immediate next step.`

6 Conclusions and Future Work

6.1 Conclusions

This project set out to design, simulate, and prototype a DC–DC boost converter capable of delivering a regulated 12 V rail from a 3S lithium polymer battery for a mobile robotic platform. The design, based on the LTC1871 current-mode controller, meets all eight performance targets in simulation with substantial margin. Output voltage regulation holds within 2 mV (0.016 %) across the full 3.1 V input swing, confirming the effectiveness of the current-mode inner loop in rejecting battery discharge dynamics. Load transient droop remains below 218 mV (1.79 %) under a 1.5 A inrush step, with recovery consistently under 50 μ s. Simulated efficiency exceeds 85 % at all operating points, ranging from 86.3 % at $V_{IN} = 8.0$ V to 89.5 % at $V_{IN} = 11.0$ V. The three-layer input protection scheme (PMOS reverse polarity blocking, resistive divider UVLO, and Zener-NPN OVLO clamp) correctly restricts active regulation to the validated 8.0–11.1 V operating window, covering the full battery discharge range. The structured three-phase methodology — analytical design, simulation verification, and iterative protection tuning — proved effective for developing a functional converter from first principles within a single academic year.

Several aspects of the design approach have value beyond this specific project. The Zener-NPN overvoltage clamp provides a low-cost, low-component-count alternative to dedicated voltage supervisor ICs such as the LTC2912, at the expense of reduced threshold accuracy — a trade-off that is acceptable for battery-powered applications where the operating window is relatively wide. The iterative tuning process required to achieve the final protection thresholds — diagnosing parasitic loading on the high-impedance UVLO divider, scaling the divider by 10 $\times$ to reduce susceptibility, and adjusting R_B to set the OVLO trip point — demonstrates a systematic debugging methodology transferable to other converter protection designs. The current balance diagnostic technique developed to investigate the efficiency anomaly (Section 5.5) successfully isolated the macro model artefact from real component losses, confirming that only 48 mW of the total loss was consumed by the controller and protection network. This technique is applicable to any simulation-based power electronics project where behavioural models obscure internal power dissipation. The topology pivot from SEPIC to boost, driven by

simulation convergence failure and validated by sufficiency analysis, illustrates that recognising when to abandon a failing approach is as important as the original design decision.

The V1 hardware prototype validated the physical design feasibility of the converter on a compact 60.5×42.5 mm two-layer PCB, but also exposed three practical engineering challenges that simulation cannot predict. First, component substitution errors by the outsourced assembly house required post-assembly verification and rework — a reminder that manufactured hardware must be inspected against the BOM before power is applied. Second, the permanent destruction of the LTC1871 controller during oscilloscope probing of the MSOP-10 package (0.5 mm pin pitch) demonstrated the risk of manual probing on fine-pitch ICs without dedicated test infrastructure. Third, the prototyping headers (J3–J5) included in the PCB design were validated as the correct approach for safe measurement access, and the experience motivates the addition of dedicated test points for critical IC signals in the V2 revision. These findings underscore the fundamental limitation of a simulation-only validation and the necessity of hardware bench testing to complete the design cycle.

6.2 Limitations

The following limitations should be considered when interpreting the results presented in this report. All eight-performance metrics and protection thresholds were validated exclusively through LTspice transient simulation; no independent electrical bench measurements were obtained due to the IC failure described in Section 5.7. Loop stability was verified analytically using a MATLAB transfer function model, not by an LTspice AC sweep or a hardware measured Bode plot. The simulated efficiency figures include a 1.39 W artefact attributed to the LTC1871 behavioural macro model (Section 5.5), meaning the reported 86–90 % efficiency range does not represent physically realisable hardware performance with full accuracy. Thermal behaviour was not characterised — no junction temperature measurements or thermal imaging were performed under sustained load. Electromagnetic interference (EMI) was not evaluated; conducted and radiated emissions from the 350 kHz switching converter remain unmeasured. The converter was tested only with an ideal pulsed current source; real downstream loads (Raspberry Pi CM4, RPLiDAR A1, RFD900x modem through point-of-load regulators) were not connected. The V1 prototype lacks the OVLO protection circuit and has a permanently damaged controller IC, leaving only two of the three protection layers physically validated. Finally, all passive components were simulated at nominal values with no Monte Carlo tolerance analysis to assess the impact of component spread on output voltage, ripple, and protection thresholds.

6.3 Future Work

The most immediate priority is to restore hardware test capability. This requires either replacing the damaged U1 (LTC1871EMS) on the existing V1 board, or fabricating the V2 PCB revision which incorporates the complete OVLO subcircuit (D3, Q4, R7–R9), the SS14 bypass Schottky in an SMA package, the corrected R8 value of 5.38 k Ω , the output bulk electrolytic C5, and dedicated test points for INTVCC, GATE, SENSE, ITH, and FB routed to accessible pads away from the IC footprint. A JLCPCB bill of materials and component placement file have already been prepared for the V2 revision. Once a functional board is available, the first bench measurements should include output voltage at three input voltages (8.0 V, 9.6 V, and 11.0 V), output ripple via oscilloscope, and input/output power for efficiency calculation. These three measurements alone would provide independent validation of the simulation predictions and close the most significant gap in this project.

In the short term, a full electrical characterisation should be conducted on the V2 board. This includes capturing the load transient response (0.5 A to 2.0 A step) on an oscilloscope to measure droop and recovery time, sweeping V_{IN} to determine the exact UVLO and OVLO thresholds on hardware, and comparing all eight-performance metrics against the simulation predictions presented in Table 5.8. Any discrepancies between simulation and hardware should be analysed and attributed to specific causes (PCB trace resistance, real component tolerances, capacitor DC bias derating, or MOSFET switching behaviour). Thermal imaging under sustained 2 A load should be performed to verify that junction temperatures of Q1 (power MOSFET), D1 (Schottky diode), and Q2 (PMOS) remain within their rated limits.

Medium-term work should address EMI compliance and system integration. A conducted emissions scan on both the input and output rails would characterise the switching noise spectrum and determine whether additional filtering (e.g., a common-mode choke or snubber network) is required. The converter should then be integrated into the robotic platform with the actual downstream point-of-load regulators and payload devices (Raspberry Pi CM4, RPLiDAR A1, RFD900x modem) to validate performance under realistic, dynamic load profiles including motor startup transients and sensor duty-cycling.

Longer-term improvements to the converter design could include replacing the Schottky diode D1 with a synchronous rectification MOSFET to recover the estimated 0.7 W diode conduction loss and push hardware efficiency above 90 %. A Monte Carlo tolerance analysis in LTspice, varying all passive component values within their specified tolerances, would quantify the statistical spread of output voltage, ripple, and protection thresholds across a production population. Finally, accelerated life testing under thermal cycling and vibration would assess the

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long-term reliability of the solder joints and component attachments for field deployment on a mobile robotic platform.

Risk Assessment

4th Year Project – Health and Safety Risk Assessment and Guidelines.

Every student must perform a health and safety risk assessment of their FYP before commencing any work.

This should be done in consultation with their project supervisor. The FYP Risk Assessment form must be assessed, along with any required material such as Material Safety Data Sheets and signed off by a member of the Health and Safety Committee.

The form is designed to identify any hazards that the proposed project might present to the student or people that may encounter the project.

Issues to consider include

- **Hazardous materials, for example flammable materials, epoxies, cements, paints, aerosols, carcinogens, poisons, toxins...**
- **Working alone, at height, in confined space, in a dangerous area**
- **Working with electricity**
- **Working with machines and tools**
- **Working with bio materials**
- **Working with heat or flame (e.g. soldering or hot air guns)**
- **Working with lasers**
- **Working with heavy or large items**
- **Working with fast spinning objects (e.g. fans or propellers)**
- **Disposable / Recycling of hazardous materials eg batteries, plastics, metal and those mentioned above.**

Where risks are identified, the student must discuss how the risk will be managed by listing what controls are in place and what further controls may be required.

The Risk Assessment Form should be included in your project report.

The Risk Assessment Form should be available for when requested by any staff member.

If this form is NOT available, then all work on the hardware aspects of your project MUST STOP until the form is signed off.

H&S committee members are: Mr. Andrew Meehan, Mr. John Maloco, Ms. Ann Dempsey, and Dr Bryan Hennelly.

I confirm, I have no medical condition that impedes working on my FYP Project, and I will advise a Health & Safety Officer if this status changes.

Signed: _____

4th YEAR PROJECT RISK ASSESSMENT FORM Rev. 3

NAME AND STUDENT NUMBER: Fortune Olose 21455956		PROJECT NAME: LTC1871-Based DC-DC Boost Converter for Battery-Powered Robotic Subsystems
SUPERVISOR: Dr. Bob Lawlor		PROJECT LOCATION: Electronics Laboratory, Maynooth University
BRIEF DESCRIPTION OF PROJECT: Design, simulation, and hardware prototyping of a DC-DC boost converter based on the LTC1871 current-mode controller. The converter accepts an 8-11 V input from a 3S lithium polymer battery and regulates a stable 12 V output at up to 2 A for a mobile robotic platform. Work involves circuit simulation in LTspice, schematic capture and PCB layout in KiCad, hand-soldering of surface-mount and through-hole components on a fabricated PCB, and electrical bench testing using a DC power supply and oscilloscope.		
Hazards, Risk [High(H) Medium (M) Low (L)], and Control Measures		
HAZARD	Risk	Controls
Electrical shock from bench power supply (0-15 V DC, 3 A) during prototype testing	L	Supply limited to 15 V DC (below 50 V safety threshold). Current-limited output. Powered off before making or breaking connections. Supervisor briefing completed.
Burns from soldering iron during hand-assembly of PCB components (including MSOP-10 fine-pitch IC)	L	Soldering performed in designated electronics laboratory under supervision of Mr. John Maloco. Temperature-controlled iron used. Heat-resistant mat and fume extraction in place. Safety glasses worn.
Inhalation of solder fumes (flux vapour) during hand-soldering and rework	L	Lead-free solder used (Sn96.5/Ag3.0/Cu0.5). Fume extraction fan positioned at workstation. Laboratory has general ventilation. Soldering sessions kept to under 2 hours.

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Electrostatic discharge (ESD) damage to sensitive CMOS IC (LTC1871EMS, MSOP-10 package)	L	ESD wrist strap worn when handling bare PCBs and ICs. Components stored in anti-static bags until assembly. ESD-safe workstation mat used in electronics laboratory.
Short circuit or component failure during power-up of prototype (thermal runaway, smoke)	L	Bench supply current limit set to 0.5 A for initial power-up. Visual inspection under magnification before first power-on. No unattended powered operation. Fire extinguisher available in laboratory.
Eye strain from prolonged use of computer for CAD (KiCad), simulation (LTspice), and report writing	L	Regular breaks taken (20-20-20 rule). Monitor positioned at correct height and distance. Laboratory and office adequately lit.
Cuts or minor injury from hand tools (side cutters, tweezers, scalpel) during PCB preparation and component trimming	L	Sharp tools used only when necessary and stored safely after use. Cut away from body. First aid kit available in laboratory.
Lithium polymer battery hazard (fire, chemical burn) if used during future integration testing	L	Battery not used in current project phase (bench supply used instead). If future testing requires battery, charge/discharge performed in fire-safe bag. Battery stored in cool, dry location. Damaged or swollen cells disposed of via university hazardous waste.
Identified risks should be discussed with your supervisor and a safe system of work agreed. A more in-depth risk assessment may be required after initial review. Do not proceed until form is signed off.		
WEEE Compliance & Recycling: Indicate below how you intend to manage and dispose of your project's material during and post project work. All electronic components and PCB assemblies produced during this project will be disposed of in accordance with the WEEE (Waste Electrical and Electronic Equipment) Directive. Populated PCBs and defective components (including the damaged LTC1871 IC) will be deposited in the designated WEEE collection bin in the Electronics Laboratory. Solder waste and flux residue will be collected and disposed of via the university's chemical waste stream. Packaging materials (anti-static bags, component reels, cardboard) will be recycled via standard campus recycling facilities. No lithium polymer batteries were used during the current project phase; if batteries are introduced in future work, they will be disposed of through the university's hazardous waste collection service and never placed in general waste.		
Further Controls Required		
No further controls are required beyond those listed above. All laboratory work was conducted under the supervision of Mr. John Maloco in the designated Electronics Laboratory. If future project phases involve integration with a		

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mobile robotic platform or the use of lithium polymer batteries, a revised risk assessment should be completed before that work commences.	
SIGNATURE OF STUDENT:	DATE:
SIGNATURE OF SUPERVISOR:	DATE:
DEPT HEALTH AND SAFETY COMMITTEE MEMBER:	DATE:

Appendix A — Generative AI Usage Declaration

In accordance with Maynooth University’s guidelines on the use of Generative AI in academic work (Section 1.6 of the EE499 FYP Handbook), this appendix documents all instances where GenAI tools were used during the preparation of this project and report.

Tool Used

Claude (Anthropic), accessed via claude.ai. Claude was used as a technical assistant throughout the project for the categories listed below. All simulation data, circuit design decisions, and analytical calculations were independently verified by the author against LTspice outputs, MATLAB results, datasheets, and first-principles theory. No GenAI-generated content was submitted without verification and adaptation by the author.

Usage Categories

- Report Writing — modifying and formatting of report sections, tables, and figure captions in Word (.docx) format.
- Simulation Analysis — interpretation of LTspice results, diagnosis of anomalies (ripple discontinuity, macro model artefact), and loss breakdown analysis.

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- Circuit Design — analytical calculations for protection circuit thresholds, compensation network values, and component selection.
- PCB and Manufacturing — BOM/CPL file generation for JLCPCB, LCSC part number verification, and package compatibility checks.
- Hardware Diagnostics — systematic fault isolation during bench testing (INTVCC anomaly, UVLO divider verification).
- Document Formatting — page count optimisation, cross-reference correction, and references compilation.
- Project Administration — email drafting, risk assessment preparation.

Representative Prompt Log

Table A.1 lists representative prompts from each usage category. The prompts shown are paraphrased summaries of the actual queries; full conversation transcripts are available on request.

#	Category	Prompt (paraphrased)	Output / Use
1	Report Writing	Format my steady-state simulation results (VOUT, ripple, droop, efficiency, recovery at four VIN points) into a structured table for Section 5.1, and refine the accompanying interpretive commentary.	Formatted Word document section with Table 5.1 containing simulation data at VIN = 8.0, 9.6, 11.0, 11.1 V, plus analytical paragraphs interpreting ripple trends, regulation, and efficiency.
2	Report Writing	Restructure my Chapter 1 draft into the required sub-sections (problem statement, methodology, evaluation strategy, broader impact, original contributions, report structure) and tighten the technical language.	Complete introductory chapter with six sub-sections, cascaded power architecture argument, and quantitative efficiency comparison (2.7 W vs 6 W).

#	Category	Prompt (paraphrased)	Output / Use
3	Report Writing	Organise my conclusions, limitations, and future work notes into a formal Chapter 6 structure with three conclusion paragraphs, a limitations section, and four-tier future work priorities.	Three-section chapter covering simulation validation, broader design value, hardware lessons, seven limitations, and prioritised future work (immediate through longer-term).
4	Report Writing	Integrate my V1 hardware notes (bench measurements, IC failure observations, V2 schematic updates) into a structured Section 5.7 with diagnostic tables and formatted commentary.	Complete section with PCB photo, bench measurement table (Table 5.10), UVLO divider verification table (Table 5.11), diagnostic analysis of INTVCC anomaly, and IC failure narrative.
5	Report Writing	Identify verbose paragraphs in Chapter 2 that can be condensed without losing technical content. Reduce word count from 3,000 to approximately 2,600 words while preserving all equations.	Identification of 13 paragraphs for trimming with replacement text. Equation-containing paragraphs were identified and excluded from modification.
6	Simulation Analysis	Analyse the OVLO circuit behaviour: with a 10 V Zener and $R_B = 5.38 \text{ k}\Omega$, calculate the expected trip voltage and verify against the LTspice sweep results at 11.0 V, 11.1 V, and 12.0 V.	Analytical calculation of $V_{\text{trip}} = V_Z + V_{BE} \times (1 + R_Z/R_B)$, with comparison to six-point simulation sweep confirming 8.0–11.1 V operating window.
7	Simulation Analysis	The loss breakdown shows 1.39 W unaccounted. The total input current is 2.844 A and inductor current is 2.839 A. Diagnose where the power is going.	Current balance diagnostic confirming only 5 mA (48 mW) drawn by IC and protection network. Residual attributed to LTC1871 behavioural macro

#	Category	Prompt (paraphrased)	Output / Use
			model overestimating internal switching losses.
8	Simulation Analysis	I've identified a 1.39 W residual in the loss breakdown that I cannot attribute to any measured component. I suspect the LTC1871 macro model is overestimating internal gate drive losses. Help me verify this hypothesis by comparing the analytical gate charge loss ($Q_G \times V_{INTVCC} \times f_{SW}$) to the residual.	Analysis linking the discontinuity to possible Zener leakage current near the 10 V breakdown voltage modulating the RUN/SS pin as V_{IN} approaches D3's breakdown region.
9	Circuit Design	Design an overvoltage lockout circuit using a Zener diode and NPN transistor that trips at approximately 11.1 V. The initial window was 8.8–10.7 V due to parasitic loading.	Iterative design process: diagnosis of parasitic loading on high-impedance UVLO divider, $10\times$ impedance scaling, selection of BZX84C10V Zener, and R_B tuning to 5.38 k Ω to achieve 8.0–11.1 V window.
10	Circuit Design	Verify my OVLO circuit design: I'm using a BZX84C10V Zener and MMBT3904 NPN to trip at approximately 11.1 V. The initial window was compressed to 8.8–10.7 V due to parasitic loading on the UVLO divider. Check my R_B calculation of 5.38 k Ω and confirm the corrected operating window.	Component values: $R_Z = 8.66$ k Ω , $C_Z = 6.8$ nF, $C_P = 100$ pF. Zero at 2.70 kHz, pole at 184 kHz. MATLAB Bode analysis confirming PM = 55.5°, GM = 19.1 dB.
11	PCB and Manufacturing	Reformat my KiCad-exported BOM into JLCPCB-compatible CSV	30-row BOM CSV with LCSC part numbers for all SMD components.

#	Category	Prompt (paraphrased)	Output / Use
		format with correct LCSC part numbers, grouped designators, and hand-solder exclusions identified.	Through-hole parts (D1, L1, C1, C5, connectors) excluded for hand-soldering.
12	PCB and Manufacturing	Reformat my KiCad-exported CPL file to match JLCPCB's required format: correct column headers, add mm suffix to coordinates, capitalise layer names, convert negative rotations.	29-component CPL CSV matching JLCPCB sample format with columns: Designator, Mid X, Mid Y, Layer, Rotation.
13	PCB and Manufacturing	C1738 is listed as the LCSC part for C2 (6.8 nF) but describes a 3.3 nF capacitor. Identify the correct part.	Identified C1755 (0805B682K500NT, 6.8 nF, 50 V, X7R, 0805) as the correct replacement. Flagged that using 3.3 nF would shift the compensation zero from 2.7 kHz to 5.5 kHz.
14	PCB and Manufacturing	D2 (1N5818) is in a DO-41 through-hole package that is oversized for the 60.5 × 42.5 mm board. Recommend an SMD alternative.	SS14 Schottky (1 A, 40 V, SMA/DO-214AC package) identified as drop-in replacement. LCSC C2480, same electrical function, compatible with automated SMD assembly.
15	Hardware Diagnostics	I'm measuring VOUT = 11.0 V from an 11.1 V input. The converter is not switching. What should I probe next?	Systematic diagnostic sequence: check INTVCC (should be 5.2 V), then FREQ, GATE, SENSE, FB, ITH pins. Priority ordered by likelihood of fault.

#	Category	Prompt (paraphrased)	Output / Use
16	Hardware Diagnostics	I'm debugging the V1 prototype — VOUT shows 11.0 V passive feed-through at VIN = 11.1 V, indicating the converter is not switching. I plan to check INTVCC first, then GATE. Confirm this is the correct diagnostic order and suggest what each reading would indicate.	Diagnosis that the internal LDO is not regulating. Three possible causes identified: defective IC, solder bridge on MSOP-10, or wrong/missing C4 bypass capacitor. Causal chain: INTVCC fault → no gate drive → no switching → passive feed-through.
17	Hardware Diagnostics	INTVCC reads 6.94 V instead of the expected 5.2 V from the internal LDO. I believe the LDO is not regulating and the pin is tracking VIN through a low-impedance path. What are the likely root causes — defective IC, solder bridge, or wrong bypass capacitor?	Back-calculation: $R6 = 2.392 \times 549k / (11.1 - 2.392) = 150.8 \text{ k}\Omega \approx 150 \text{ k}\Omega$ standard value. Deviation from predicted 2.382 V is 0.42 %.
19	Document Formatting	Scan the document for broken cross-references caused by the removal of sub-subsection headings, and suggest corrected section numbers.	Systematic scan identifying all sub-section references. Replacements applied: Section 2.2.2 → Section 2.2, Section 4.6 → Section 4.6, Chapter 5 → Chapter 5.
20	Document Formatting	Format my list of 23 source citations into IEEE style with hanging indents, and verify that each [X] marker in the report body maps to the correct reference entry.	23-entry reference list covering datasheets, textbooks, standards, software tools, and UN SDGs, formatted with hanging indents in IEEE style.

#	Category	Prompt (paraphrased)	Output / Use
21	Project Administration	Refine my draft email to Dr. Bob Lawlor requesting feedback on the current report, flagging the outstanding items (Chapter 6, references, GenAI appendix).	Two email variants (detailed and brief) flagging outstanding items: Chapter 6, references, GenAI appendix.

Table A.1 — Representative Generative AI prompts used during the EE499 project

Verification Statement

All technical content generated with AI assistance was independently verified by the author. Simulation data was cross-checked against LTspice .meas outputs and waveform plots. Analytical calculations (duty cycle, inductor sizing, compensation network, protection thresholds) were verified against datasheet specifications and first-principles equations derived in Chapter 2. Component selections were validated against manufacturer datasheets. BOM entries were verified against LCSC inventory listings before ordering. The author takes full responsibility for the accuracy and integrity of all content presented in this report.

Appendix B

Table 16 - Bill of Materials

Ref	Description	Part Number	Footprint	Qty	Total (€)
U1	LTC1871 controller, MSOP-10	LTC1871EMS#PBF	MSOP-10	1	3.80
Q1	IRF7811 MOSFET, SO-8	IRF7811TRPBF	SO-8	1	1.20
D1	MBR735 Schottky diode	MBR735	TO-220	1	0.45
L1	10 μ H inductor	XAL6060-103MEC	10×10 mm SMD	1	2.80
C1a (×7)	22 μ F, 25 V, X5R MLCC	GRM31CR61E226M	1206	7	3.15
C1b (×2)	10 μ F, 25 V, X5R MLCC	GRM21BR61E106K	0805	2	0.50

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C5	100 μ F, 25 V electrolytic	Generic	Radial 8 mm	1	0.20
C _{in}	100 μ F, 25 V electrolytic	Generic	Radial 8 mm	1	0.25
R2	56 k Ω , 1 % (R _{BOT})	—	0805	1	0.02
R3	499 k Ω , 1 % (R _{TOP})	—	0805	1	0.02
R _Z	8.66 k Ω , 1 %	—	0805	1	0.02
C _Z	6.8 nF, 50 V	—	0805	1	0.03
C _P	100 pF, 50 V	—	0805	1	0.02
R _{RT}	68.1 k Ω , 1 %	—	0805	1	0.02
R6	49 k Ω , 1 % (UVLO top)	—	0805	1	0.02
R7	10 k Ω , 1 % (UVLO bottom)	—	0805	1	0.02
R5	1 k Ω (OVLO series)	—	0805	1	0.02
D2	BZX84C10 Zener	BZX84C10	SOT-23	1	0.08
Q2	BC547 NPN	BC547	SOT-23	1	0.04
D3	AO6407 (reverse protection)	AO6407	SOT-23-6	1	0.40
Total					11.30

Table 17: Measurement Directives and Windows

The complete results summary with pass/fail status for all twelve metrics is provided in Appendix B, Table B.6.

Table 18: LTspice Component Models and Library Sources

Component	Model / Library	Source	Notes
LTC1871 controller	LTC1871.sub (macro model)	Analog Devices [19]	Behavioural model; captures functional behaviour, current-mode control, protection features. May not perfectly predict

			switching losses or gate drive dynamics.
N-channel MOSFET (IRF7811)	standard.mos (NMOS)	LTspice built-in	Level 1 model with parameters extracted from Vishay datasheet.
P-channel MOSFET (AO6407 / FDS4435A)	standard.mos (PMOS)	LTspice built-in	Used for reverse polarity protection.
Schottky diode (MBR735)	standard.dio	LTspice built-in	Includes forward voltage drop and junction capacitance.
Zener diode (BZX84C10VLY)	standard.dio (zeners)	LTspice built-in	10 V breakdown voltage.
NPN transistor (OVLO)	standard.bjt	LTspice built-in	Used MMBT3904 model (corresponds to OVLO circuit in validated netlist).
Inductor L1	behavioural	Ideal with $R_{ser}=14m$	10 μH , 14 m Ω series resistance (typical DCR).
Capacitors C1a, C1b, C5	behavioural	Ideal with series resistance as shown	Derated capacitance values (77 μF , 12 μF , 100 μF).
Input voltage source V1	behavioural	{VIN} parameterised	Allows parametric step. startup flag enabled.
Load current source I1	behavioural	PWL function	Described in Section 4.3.

Table B.4 — Efficiency and Power Balance Across Input Range

The complete efficiency and power balance data across the input voltage range is provided in Appendix B, Table B.4.

Table B.5 — Current Balance Diagnostic

The current balance diagnostic measurements are provided in Appendix B, Table B.5.

Table B.6 — Complete Results Summary

The complete results summary with pass/fail status for all twelve metrics is provided in Appendix B, Table B.6.

Table B.7 — V1 Component Cross-Reference (KiCad/LTspice)

The complete V1 prototype component cross-reference (KiCad and LTspice designators, population status) is provided in Appendix B, Table B.7.

Table B.6 — Complete Results Summary

Metric	.meas Directive	Window	Rationale
Pre-step output voltage	AVG V(OUT) FROM 4m TO 5m	4–5 ms	After startup settles (by 2 ms), before load step (5 ms).
Post-step output voltage	AVG V(OUT) FROM 6m TO 7m	6–7 ms	After load step transient has fully recovered.
Load transient droop (voltage)	MIN V(OUT) FROM 5.1m TO 5.4m	5.1–5.4 ms	Captures minimum value during the 100 μ s ramp and subsequent decay.
Droop in millivolts	PARAM (Vout_before – Vout_droop)*1000	Derived	Calculated from captured pre-step average and minimum.
Droop as percentage	PARAM (Vout_before – Vout_droop)/Vout_before*100	Derived	Expresses droop relative to nominal output.
Output ripple (peak-peak)	PP V(OUT) FROM 6m TO 7m	6–7 ms	Steady-state at full load, after recovery.
Ripple in millivolts	PARAM Vout_ripple*1000	Derived	Converts peak-peak ripple to millivolts for readability.
Peak inductor current	MAX I(L1) FROM 4m TO 5m	4–5 ms	Pre-step, light load (0.5 A) – used for saturation check.
Recovery time	FIND TIME WHEN V(OUT)=12.15 CROSS=2 TD=5m	After 5 ms	Time at which output returns to 12.15 V ($\approx$ 99% of target) after the step.
Input power	AVG V(IN)*(-I(V1)) FROM 6m TO 7m	6–7 ms	Post-step, full load, steady-state.

Metric	.meas Directive	Window	Rationale
Output power	AVG V(OUT)*I(I1) FROM 6m TO 7m	6–7 ms	Post-step, full load, steady-state.
Efficiency	PARAM Pout/Pin*100	Derived	Calculated after both power measurements.

Table B.4 — Efficiency and Power Balance

VIN (V)	Pin (W)	Pout (W)	Ploss (W)	η (%)
8.0	28.248	24.390	3.858	86.3
9.6	27.305	24.395	2.909	89.3
11.0	27.247	24.396	2.851	89.5
11.1	27.354	24.395	2.960	89.2

Table B.5 — Current Balance Diagnostic

Measurement	Value	Significance
Total input current (from V1)	2.844 A	All current drawn from battery
Average inductor current (I_L1)	2.839 A	Current in power path
Difference (IC + protection)	0.005 A (48 mW)	Negligible controller draw
OVLO R9 current	$\approx 10^{-11}$ A	Zener fully off at 9.6 V
OVLO R10 current	$\approx 10^{-11}$ A	NPN base fully off
Input capacitor RMS current	68 μ A	No measurable ESR loss

Table B.6 — Complete Results Summary

Metric	Worst-Case Result	Design Target	Margin	Status
Output voltage	12.196–12.198 V	12.0 V ($\pm 1\%$)	+1.6%	✓ PASS
Line regulation	2 mV / 3.1 V	< 1% / V	33,000×	✓ PASS
Output ripple (pk-pk)	177.8 mV (1.46%)	< 240 mV (2%)	1.35×	✓ PASS
Load transient droop	217.8 mV (1.79%)	< 600 mV (5%)	2.7×	✓ PASS

Metric	Worst-Case Result	Design Target	Margin	Status
Recovery time	50 μ s	< 1 ms	20×	✓ PASS
Peak inductor current	3.561 A	< 11.2 A (Isat)	3.1×	✓ PASS
Efficiency (simulated)	86.3%	> 85%	+1.3 pp	✓ PASS
Phase margin	55.5°	> 45°	+10.5°	✓ PASS
Gain margin	19.1 dB	> 10 dB	+9.1 dB	✓ PASS
UVLO turn-on	8.0 V	$\leq$ 8 V	At boundary	✓ PASS
OVLO trip	11.1–12.0 V	$\geq$ 11 V	At boundary	✓ PASS
Operating window	8.0–11.1 V	8–11 V	3.1 V band	✓ PASS

Table B.7 — V1 Component Cross-Reference

KiCad Ref	LTspice Ref	Component	V1 Status	Function
U1	U1	LTC1871EMS (MSOP-10)	Present	Controller
Q1	M1	SI4894BDY (SO-8)	Present	Power MOSFET
Q2	M2	DMP3099L (SOT-23)	Present	Reverse polarity PMOS
D1	D1	MBR735 (TO-220)	Present	Schottky rectifier
L1	L1	XAL6060-103MEC (10 μ H)	Present	Inductor (shielded)
R1	R1	68.1 k Ω (0805)	Present	FREQ set (350 kHz)
R2	R2	56 k Ω (0805)	Present	FB lower divider
R3	R3	499 k Ω (0805)	Present	FB upper divider
R4	R4	8.66 k Ω (0805)	Present	Compensation resistor
R5	R6	49 k Ω (0805)	Present	UVLO upper divider
R6	R7	10 k Ω (0805)	Present	UVLO lower divider

KiCad Ref	LTspice Ref	Component	V1 Status	Function
C1	Cin	100 μ F electrolytic	Present	Input bulk capacitor
C2	C2	6.8 nF (0805)	Present	Compensation (ITH zero)
C3	C3	100 pF (0805)	Present	Compensation (ITH pole)
C4	C4	4.7 μ F (0805)	Present	INTVCC bypass
C6	—	22 μ F MLCC (1206)	Present	Input HF decoupling
C7	—	22 μ F MLCC (1206)	Present	Input HF decoupling
C1a1–C1a7	C1a	7 $\times$ 22 μ F MLCC (1206)	Present	Output MLCC bank
C1b1–C1b2	C1b	2 $\times$ 10 μ F MLCC (0805)	Present	Output MLCC bank
J2	—	Output connector	Present	Load connection
J3–J5	—	Pin headers	Present	Debug access
D2	D2	SS14 Schottky (SMA)	Missing	PMOS body diode bypass
D3	D3	BZX84C10V Zener (SOT-23)	Missing	OVLO clamp
Q4	Q1 (LTspice)	MMBT3904 NPN (SOT-23)	Missing	OVLO clamp transistor
R7	R8	10 k Ω (0805)	Missing	PMOS gate pulldown
R8	R9	5.38 k Ω (0402)	Missing	OVLO base pulldown (RB)
R9	R10	10 k Ω (0402)	Missing	Zener current limiter
C5	C5	100 μ F electrolytic	Missing	Output bulk capacitor

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